

Valence Asymmetries in Information Packaging and the Expressive Contribution of Evaluative Adjectives



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ASYMMETRIES



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(1a) *“Italy’s boring Olympics have been held in Milan and Cortina.”*

TRUE or **FALSE?**

(1a) *“Italy’s boring Olympics have been held in Milan and Cortina.”*

TRUE or **FALSE?**

(2a) *“Romeo and Juliet is a tragedy by Shakespeare and it’s terrific.”*

TRUE or **FALSE?**

(1a) “Italy’s *boring* Olympics have been held in Milan and Cortina.”

▶ **(1b)** Italy’s Olympics are *boring*. (**Opinion-based content**)

(2a) “Romeo and Juliet is a tragedy by Shakespeare and it’s *terrific*.”

▶ **(2b)** Romeo and Juliet is *terrific*. (**Opinion-based content**)

(1a) “Italy’s *boring* Olympics *have been held in Milan and Cortina.*”

▶ **(1b)** Italy’s Olympics are *boring*. (**Opinion-based content**)

▶ **(1c)** Italy’s Olympics *have been held in Milano and Cortina.* (**Factual content**)

(2a) “Romeo and Juliet *is a tragedy by Shakespeare and it’s terrific*”

▶ **(2b)** Romeo and Juliet is *terrific*. (**Opinion-based content**)

▶ **(2c)** Italy’s Olympics *is a tragedy by Shakespeare.* (**Factual content**)

(1a) “Italy’s *boring* Olympics *have been held in Milan and Cortina*.”

- ▶ **(1b)** Italy’s Olympics are *boring*. (**Opinion-based content**)
- ▶ **(1c)** Italy’s Olympics *have been held in Milano and Cortina*. (**Factual content**)
- ▶ **(1d)** The *negative stance* hold by the speaker towards the referent. (**Evaluative content**)

(2a) “Romeo and Juliet *is a tragedy by Shakespeare* and it’s *terrific*.”

- ▶ **(2b)** Romeo and Juliet is *terrific*. (**Opinion-based content**)
- ▶ **(2c)** Italy’s Olympics *is a tragedy by Shakespeare*. (**Factual content**)
- ▶ **(2d)** The *negative stance* hold by the speaker towards the referent. (**Evaluative content**)

Some Background

Subjectivity (Faultless Disagreement)

(1a) “Italy’s *boring* Olympics have been held in Milan and Cortina.”

- ▶ A: Italy’s Olympics are *boring* ✓
B: **No**, they’re not! ✓

$\llbracket p \rrbracket_{c\langle w,j \rangle}$	$\langle w@,j_A \rangle$	$\langle w@,j_B \rangle$
c_A	$p=1$	$/$
c_B	$/$	$p=0$

[+Disagreement,-Faultless]

(2a) “Romeo and Juliet is a tragedy by Shakespeare and it’s *terrific*.”

- ▶ A: Romeo and Juliet is *terrific* ✓
B: **No**, it’s not! ✓

$\llbracket p \rrbracket_{c\langle w,j \rangle}$	$\langle w@,j_A \rangle$	$\langle w@,j_B \rangle$
c_A	$p=1$	$/$
c_B	$/$	$p=0$

[+Disagreement,-Faultless]

(Kölbel, 2004; Lasersohn, 2004, 2009; MacFarlane, 2015)

Some Background

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(1a) “Italy’s boring Olympics *have been held in Milano and Cortina*.”

- ▶ A: Italy’s Olympics *have been held in Milano and Cortina* ✓
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$\llbracket p' \rrbracket_{cw}$	$w@$
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[+Disagreement, +Faultless]

(2a) “Romeo and Juliet *is a tragedy by Shakespeare and it’s terrific*.”

- ▶ A: Romeo and Juliet *is a tragedy by Shakespeare* ✓
B: **No**, it’s not! ✗

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[+Disagreement, +Faultless]

(Kölbel, 2004; Lasersohn, 2004, 2009; MacFarlane, 2015)

Some Background

Evaluative content (Valence)

(1a) “Italy’s *boring*^[neg-] Olympics have been held in Milano and Cortina.”



The adjective *boring* expresses the *speaker’s negative stance* towards ‘Italy’s Winter Olympics’ 😞👎.

(2a) “Romeo and Juliet is a tragedy by Shakespeare and it’s *terrific*^[pos+].”



The adjective *terrific* expresses the *speaker’s positive stance* towards ‘Romeo and Juliet’ 😊👍.

(Ruiz & Stojanovic, 2019; Umbach, 2022; Stojanovic, 2025)

Some Background

Information Packaging

(1a) “Italy’s *boring*^[neg-] Olympics *have been held in Milano and Cortina.*”

Topic/Given

Comment/New

(2a) “Romeo and Juliet *is a tragedy by Shakespeare* and it’s *terrific*^[pos+].”

Topic/Given

Comment/New

(Chafe, 1976; Reinhart, 1981; Krifka, 2008)

Some Background

Information Packaging

(1a) “Italy’s *boring*^[neg-] Olympics *have been held in Milano and Cortina.*”

Topic/Given

- [DP Italy’s Winter Olympics]
- [VP they are boring]

Comment/New

- [VP held in Milan and Cortina]

(2a) “Romeo and Juliet *is a tragedy by Shakespeare* and it’s *terrific*^[pos+].”

Topic/Given

- [DP Romeo and Juliet]

Comment/New

- [VP is a tragedy by Shakespeare]
- [VP it is terrific]

- existential presupposition
 - factual content
 - opinion-based content
- (Chafe, 1976; Reinhart, 1981; Krifka, 2008)

Some Background

Information Packaging - processing

Attention Allocation	Topic/Given	Comment/New
<i>Vigilance</i>	<i>Low</i>	<i>High</i>
<i>Egocentric bias</i>	<i>Sale resistance (low)</i>	<i>Sale resistance (high)</i>
<i>Force</i>	<i>Locutionary Force</i>	<i>Illocutionary Force</i>
<i>Dual processing</i>	<i>Automatic processing</i>	<i>Controlled Processing</i>
<i>RTs/Cog.effort</i>	<i>Faster/Lower</i>	<i>Slower/Higher</i>

Table 1. Expected cognitive processing effects as a function of the information structure of the sentence.

(Rosengren, 1992, 1993; Ferreira et al., 2002; Ward & Sturt, 2007; Sperber et al., 2010; Christiansen & Chater, 2016; Lombardi Vallauri, 2016; Domaneschi et al., 2018; Lombardi Vallauri, 2021)

Some Background

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How do comprehenders **processes**
subjective content?

Previous Studies

Kaiser and Wang (2021)

Aims of the Study

Tested whether people's ability to “**perceive**” *opinion-based content*, as indicated by the use of **subjective adjectives** (e.g., *amazing, tall*), depends on the **linguistic position** of the adjective. This was contrasted with **objective adjectives**, which are not opinion-based.



Elsi Kaiser

Previous Studies

Kaiser and Wang (2021)

Test Items

► **Baseline:**

(3a) The [$\emptyset$] orchestra included five prize-winning violinists.

► **Prenominal (PN):**

(3b) The [*amazing/huge/Viennese*] orchestra included five prize-winning violinists.

► **Appositive Relative Clause (RC):**

(3c) The orchestra, which was [*amazing/huge/Viennese*], included five prize-winning violinists.

► **Predicative (PD):**

(3d) The orchestra was [*amazing/huge/Viennese*]. It included five prize-winning violinists.

Instructions



“The amazing orchestra included five-prize winning violinists.”

Fact	1	2	3	4	5	6	Opinion
	☐	☐	☐	☐	☐	☐	

(Kaiser & Wang, 2021, p. 9)

Previous Studies

Kaiser and Wang (2021)

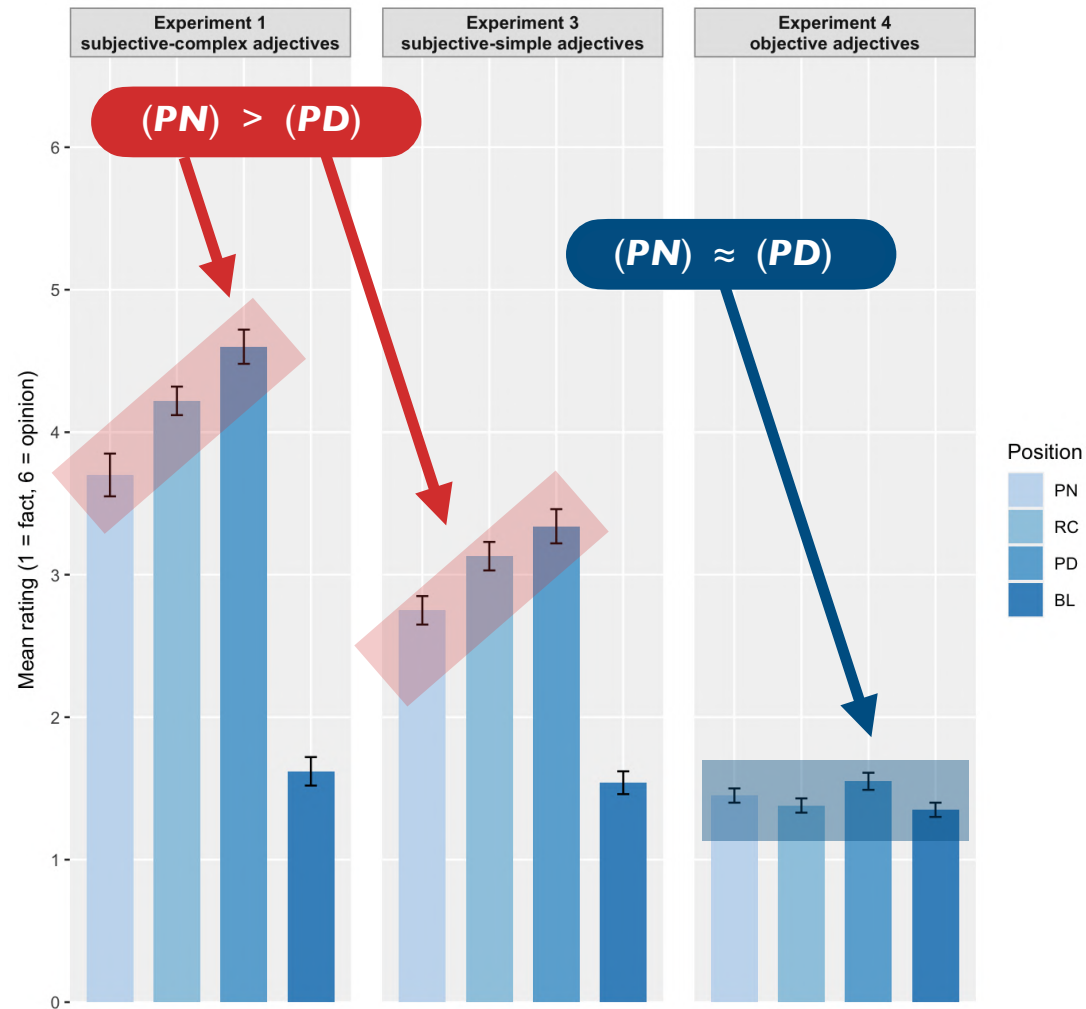


Figure 1. Experiments 1, 3, and 4 replotted from Kaiser & Wang (2021, pp. 10, 17, 20).

Previous Studies

Kaiser and Wang (2021)

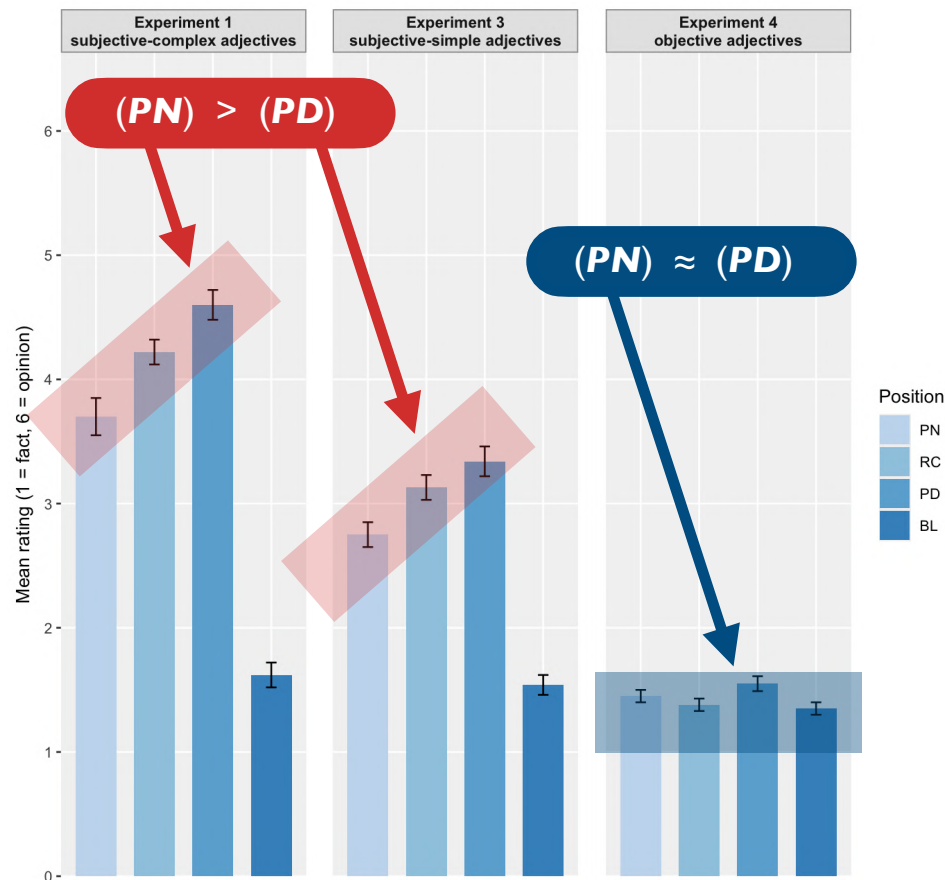


Figure 1. Experiments 1, 3, and 4 replotted from Kaiser & Wang (2021, pp. 10, 17, 20).

► Structural Effects

(i) Perceived subjectivity:

$$PD_{New} > RC > PN_{Given} \checkmark$$

► Packaging vs. Semantics

(i) Structure-based hypothesis:

$$H_{Structure} > H_{Lexical} \checkmark$$

(ii) Lexical-based hypothesis:

$$PD_{New} \approx RC \approx PN_{Given} \times$$

Previous Studies

Kaiser and Wang (2021)

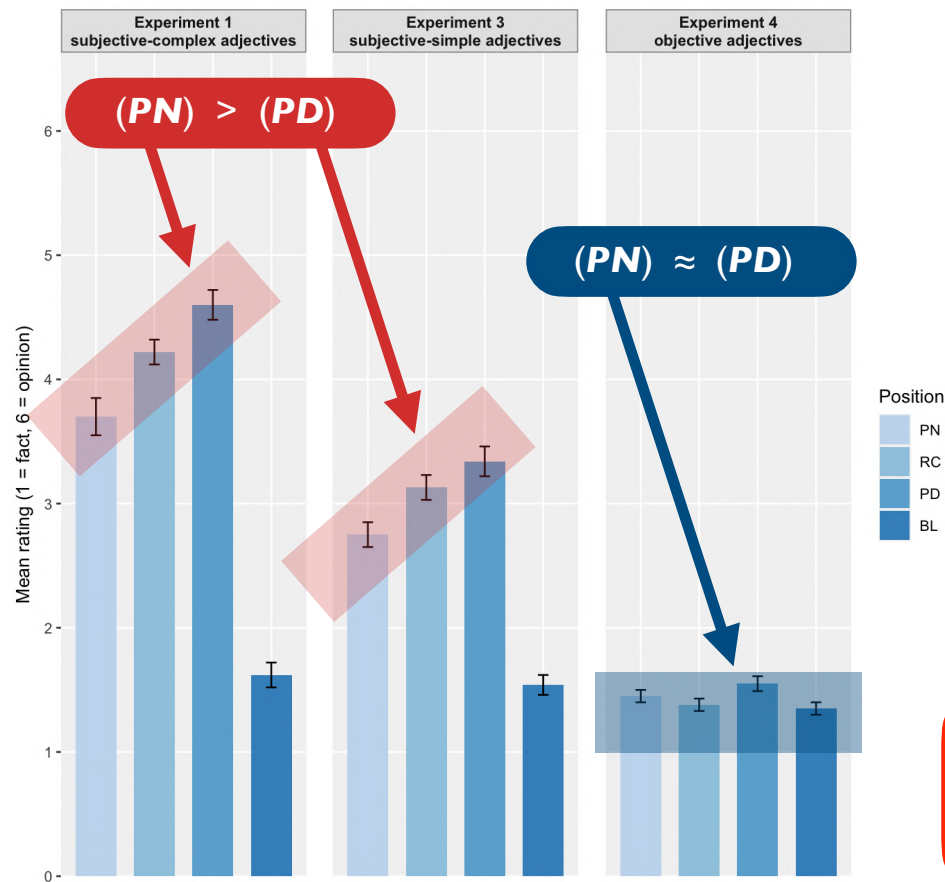


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(i) Structure-based hypothesis:

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(ii) Lexical-based hypothesis:

$$PD_{New} \approx RC \approx PN_{Given} \times$$

Takeaway!

Subjective adjectives (e.g. *amazing*, *terrible*, *tall*, *rich*) in **PN** position are “perceived as more objective” than when they occur in **RC** or in **PD** position.

Previous Studies

Delogu & Degaetano-Ortlieb (2025)

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Aims of the Study

Tested whether subjects' *plausibility* and *agreement*¹ of opinion-based content, particularly controversial opinions, depends on the linguistic packaging of the **subjective adjective** (**Given** vs. **New**), and whether this interacts with opinion type (**Controversial** vs. **Non-controversial**) and **speaker expertise**.



Francesca Delogu



Stefania Degaetano-Ortlieb

Previous Studies

Delogu & Degaetano-Ortlieb (2025)

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Test Items

▶ Given, Non-controversial:

(4a) The **brehtaking** Great Pyramids of Giza are largely intact.

▶ New, Non-controversial:

(4b) The Great Pyramids of Giza are **brehtaking** and largely intact.

▶ Given, Controversial:

(4c) The **uninteresting** Great Pyramids of Giza are largely intact.

▶ New, Controversial:

(4d) The Great Pyramids of Giza are **uninteresting** and largely intact.

Instructions



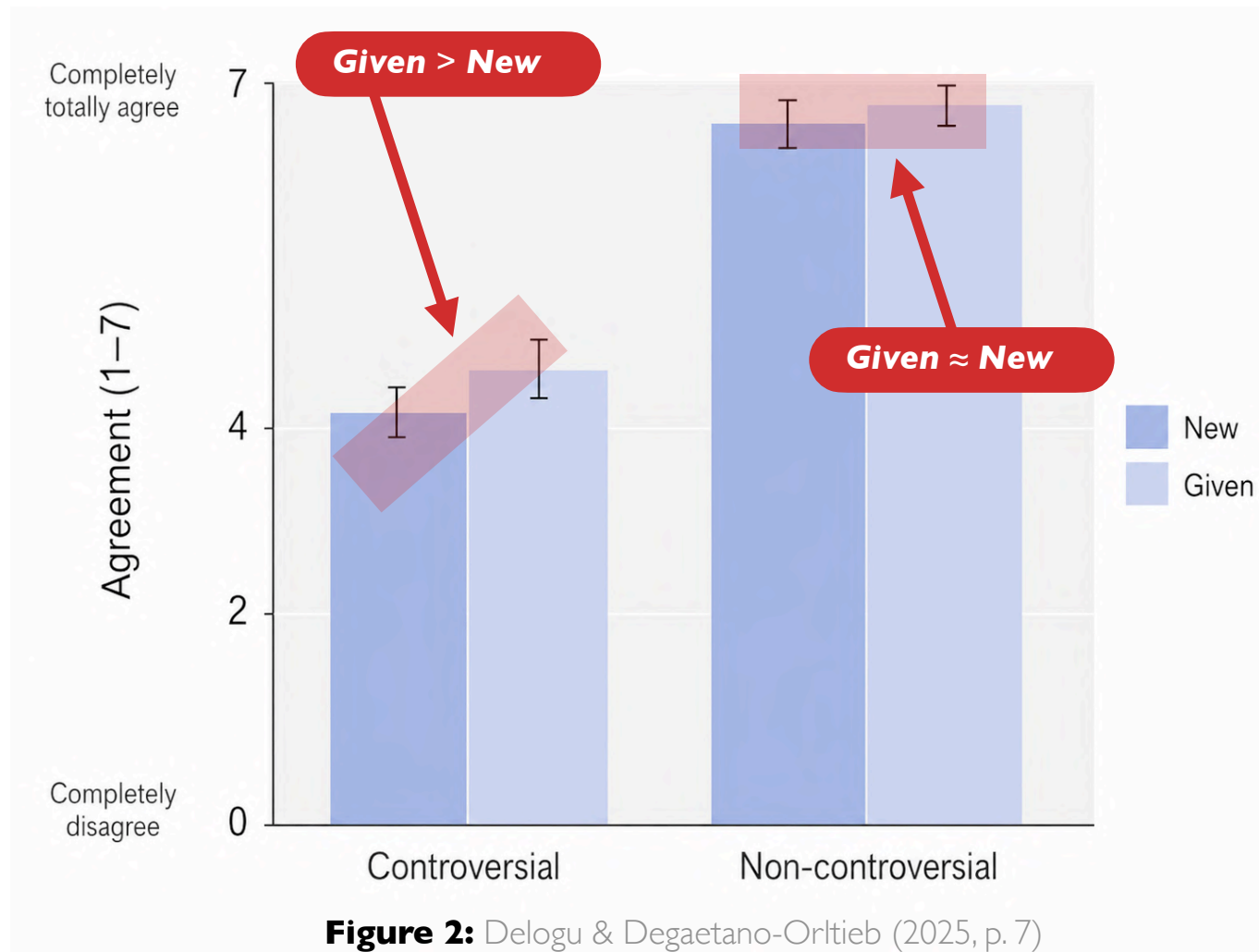
“The uninteresting Great Pyramids of Giza are largely intact.”

Totally Disagree	1	2	3	4	5	6	7	Totally Agree
	○	○	○	○	○	○	○	

(Delogu & Degaetano-Ortlieb, 2025, pp. 5,8)

Previous Studies

Delogu & Degaetano-Ortlieb (2025)



Previous Studies

Delogu & Degaetano-Ortlieb (2025)

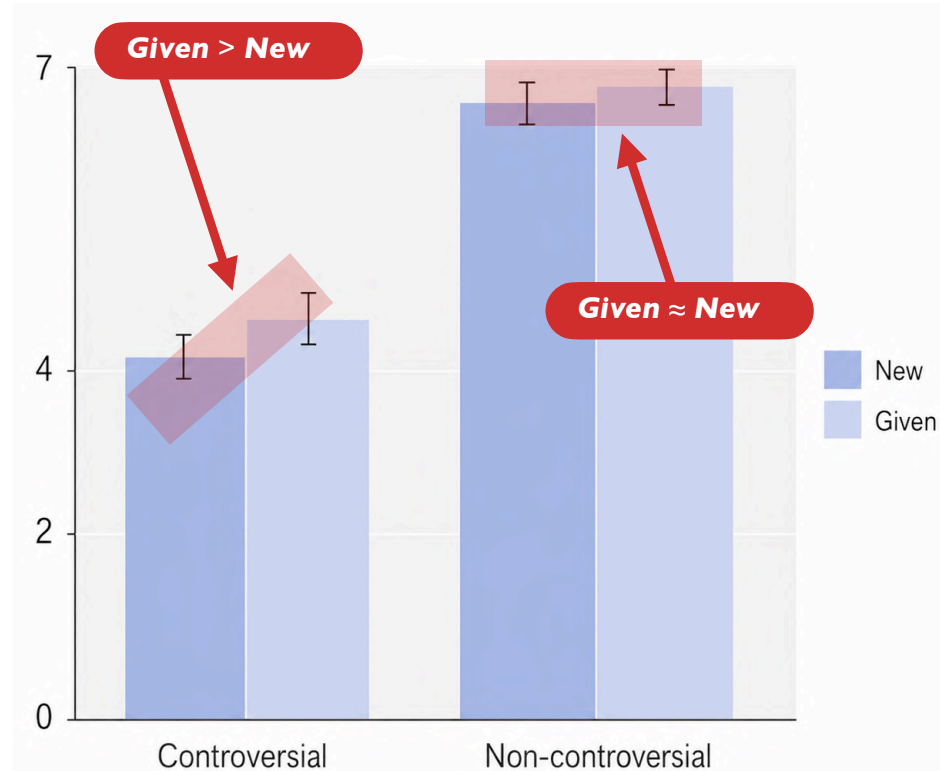


Figure 2: Delogu & Degaetano-Ortlieb (2025, p. 7)

► **OpinionType**

- (i) Agreement responses:
Non-controversial > Controversial ✓

► **OpinionType × Structure**

- (i) when controversial:
Given > New ✓
- (ii) when non-controversial:
Given ≈ New ✗

Previous Studies

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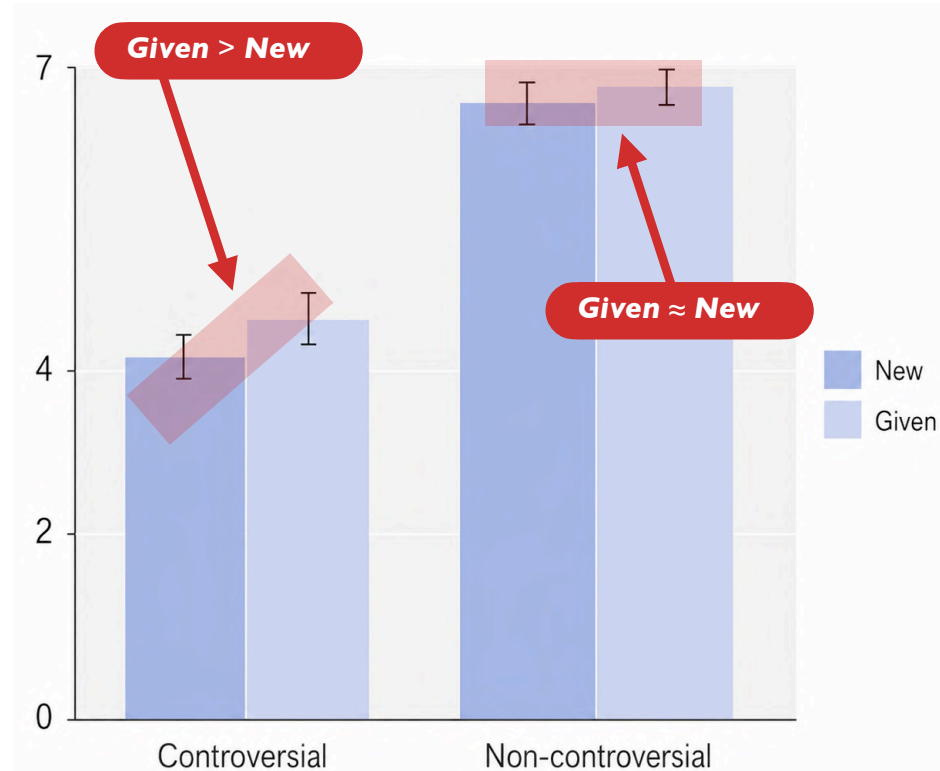


Figure 2: Delogu & Degaetano-Ortlieb (2025, p. 7)

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- (i) Agreement responses:
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► **OpinionType × Structure**

- (i) when controversial:
Given > New ✓
- (ii) when non-controversial:
Given ≈ New ✗

Takeaway!

Controversial opinions are accepted to a higher degree when they are presented as **Given (PN)** rather than **New (PD)**

Previous Studies

Delogu & Degaetano-Ortlieb (2025)

Kaiser & Wang's Assumption

“Our studies show that the power of linguistic packaging extends to the subjective domain, and can have significant effects on humans’ ability to pick up on opinion-based information: If a speaker/writer wants to *sneakily present an opinion as an objective fact*, they should present the relevant subjective information in prenominal modifier position and not in predicative position.” (Kaiser & Wang, 2021, p. 22)

Delogu & Degaetano-Ortlieb's Assumption

The present work provides evidence that even *controversial opinions*, with which most hearers would disagree, may receive *reduced epistemic vigilance* when hidden in the linguistic signal and presented as widely shared objective views. This leads people to accept these opinions to a higher degree than when they are the focus of attention.” (Delogu & Degaetano-Ortlieb, 2025, p. 15)

Previous Studies

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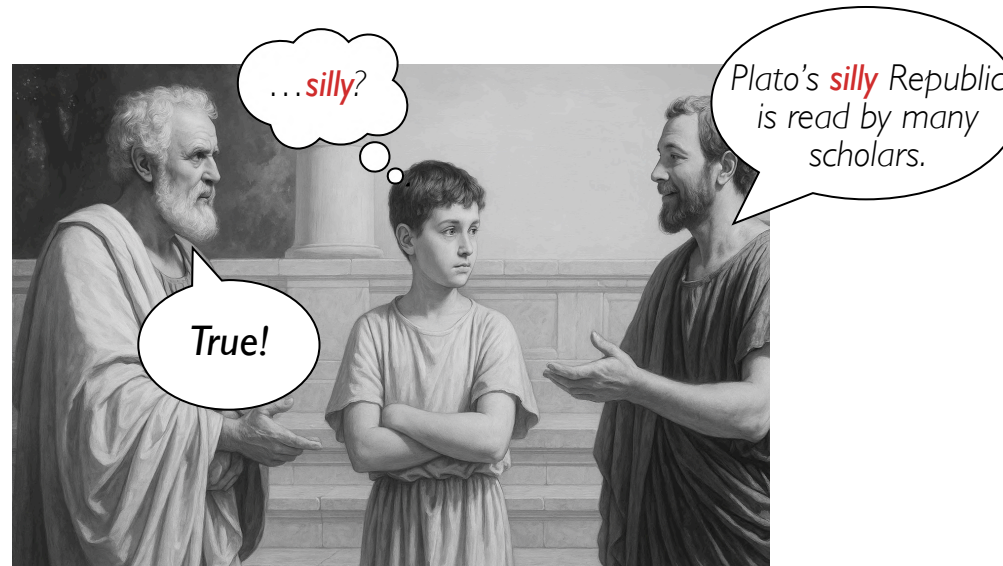
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Where to go from here?

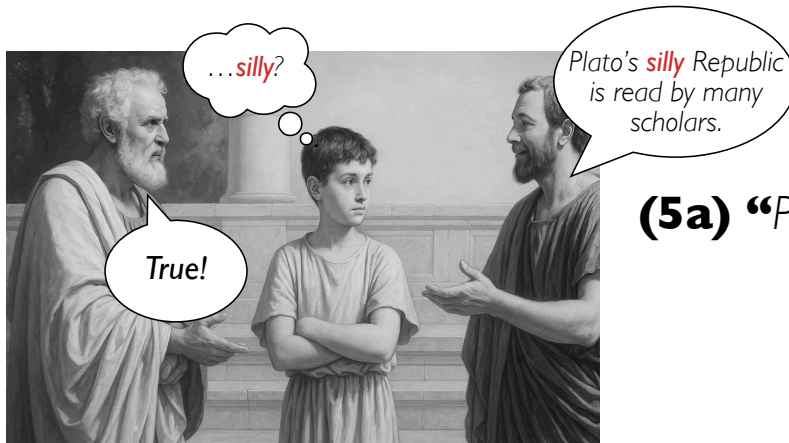
What is the Truth-Conditional Status of Subjective Content?

Following Kaiser & Wang (2021) and Delogu & Degaetano-Ortlieb (2025), the present study extends their findings by investigating whether **information packaging** also affects the semantics of **subjective content** in the form of controversial opinions.



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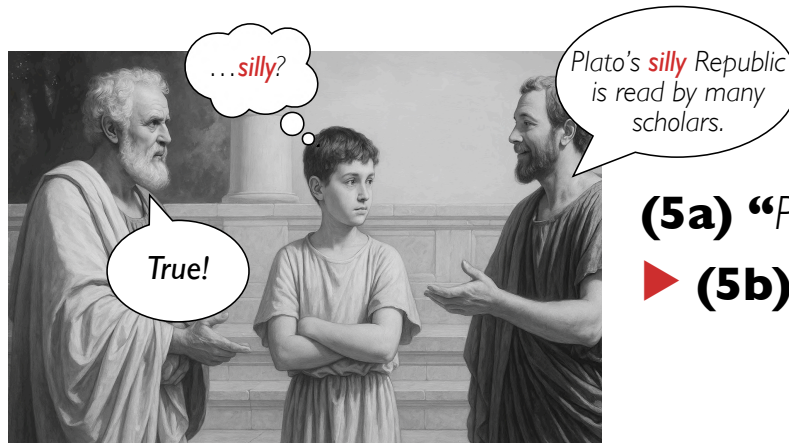
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(5a) “Plato’s *silly* Republic *is read by many scholars*.”

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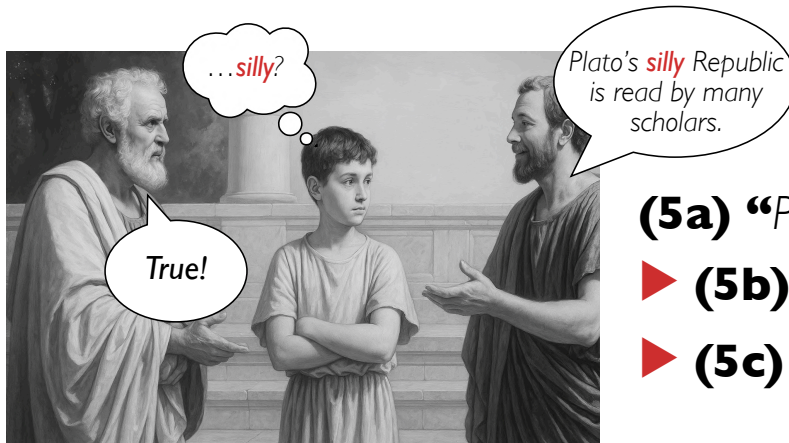


(5a) “Plato’s *silly* Republic *is read by many scholars.*”

► **(5b) New:** Plato’s Republic *is read by many scholars.*

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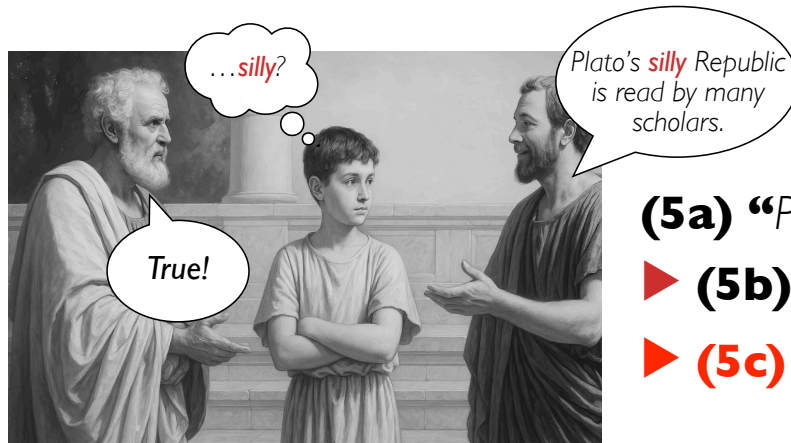
(5a) “Plato’s *silly* Republic *is read by many scholars*.”

► **(5b) New:** Plato’s Republic *is read by many scholars*.

► **(5c) Given:** Plato’s Republic is *silly*[neg].

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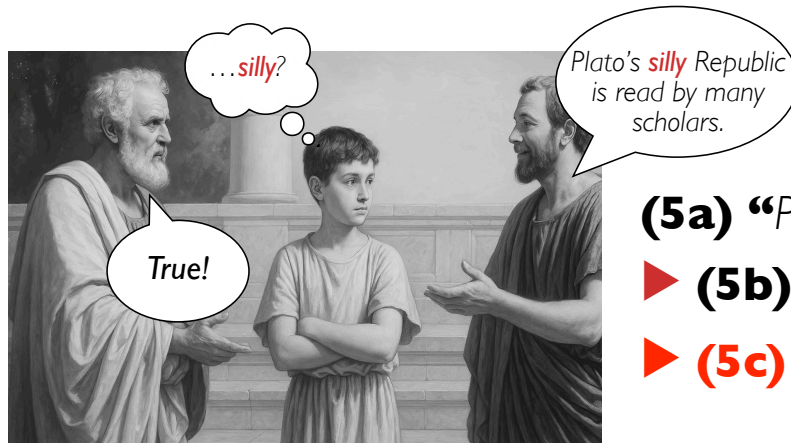
► **(5b) New:** Plato’s Republic *is read by many scholars*.

► **(5c) Given:** Plato’s Republic is *silly*[**neg**].

► **Q1.** Does **packaging** affect the truth-conditions of **opinion-based content**?

What is the Truth-Conditional Status of Subjective Content?

Following Kaiser & Wang (2021) and Delogu & Degaetano-Ortlieb (2025), the present study extends their findings by investigating whether **information packaging** also affects the semantics of **subjective content** in the form of controversial opinions.



(5a) “Plato’s *silly Republic is read by many scholars.*”

► (5b) **New:** Plato’s Republic *is read by many scholars.*

► (5c) **Given:** Plato’s Republic *is silly[neg].*

► **Q1.** Does **packaging** affect the truth-conditions of **opinion-based content**?

► **Q2.** Does **opinion-based content** get **accommodated** or **rejected**?

Design-Rationale

Stimuli design (Experiment I)

Stimuli consisted of familiar entities **(i)** presented in definite NPs, combined with positively or negatively valenced adjectives **(ii)** and true or false continuations **(iii)**.

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► Familiar entities

e.g. *Plato's work Republic, Monet's paintings, The Great Pyramids of Giza, The actor Leonardo diCaprio, Canada's maple syrup, ...*

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► Subjective adjectives: **positive vs. negative valenced**

e.g. [*inspiring/silly*], [*beautiful/ugly*], [*amazing/boring*], [*extraordinary/overrated*], [*tasty/disgusting*], ...



Adjectives	Count (%)	Example
Other	14.6% (n = 7)	French Revolution [<i>inspiring, tedious</i>]; Freud's works [<i>engaging, tiresome</i>]; Harvard University [<i>stimulating, stressful</i>]
Aesthetic	50.0% (n = 24)	Van Gogh's pieces of art [<i>beautiful, ugly</i>]; Michelangelo's works [<i>astonishing, unimpressive</i>]; Buckingham Palace [<i>stylish, dull</i>]
Taste	35.4% (n = 17)	Disney's rides [<i>amusing, boring</i>]; Belgian chocolate [<i>delicious, disgusting</i>]; Titanic [<i>touching, terrible</i>]

Table 2. Classification and distribution of the evaluative adjectives included in the study.

Truth-Judgment Studies

Design-Rationale

	Positive	Negative
1. Van Gogh's pieces of art [<i>beautiful</i> , <i>ugly</i>]	6.88	1.12
2. habit of exercising [<i>pleasant</i> , <i>unpleasant</i>]	6.88	1.25
3. ritual of eating sweets at night [<i>comforting</i> , <i>sickening</i>]	6.25	1.12
4. Michelangelo's works [<i>astounding</i> , <i>unimpressive</i>]	5.88	1.12
5. Tacos [<i>delicious</i> , <i>disgusting</i>]	6.62	2.25
6. French Revolution [<i>inspiring</i> , <i>tedious</i>]	6.25	2.25
7. Great Pyramid of Giza [<i>brehtaking</i> , <i>uninteresting</i>]	5.88	1.88
8. Neil Armstrong's moon landing [<i>thrilling</i> , <i>boring</i>]	5.50	1.62
9. actress Marilyn Monroe [<i>attractive</i> , <i>unattractive</i>]	6.12	2.38
10. movie Titanic [<i>touching</i> , <i>terrible</i>]	5.50	1.75
11. tennis player Nadal [<i>exciting</i> , <i>unpleasant</i>]	5.50	1.75
12. composer Tchaikovsky [<i>lyrical</i> , <i>jarring</i>]	5.75	2.12
13. habit of warming up before working out [<i>pleasant</i> , <i>stressful</i>]	6.38	2.88
14. Barcelona's football club FC Barcelona [<i>elegant</i> , <i>clumsy</i>]	6.12	2.62
15. Belgian chocolate [<i>delicious</i> , <i>disgusting</i>]	5.88	2.38

Table 3. Mean pre-test ratings sample for positive and negative evaluative adjectives across 48 familiar target entities; higher values indicate greater perceived plausibility.

Stimuli design (Experiment I)

Stimuli consisted of familiar entities **(i)** presented in definite NPs, combined with positively or negatively valenced adjectives **(ii)** and true or false continuations **(iii)**.

► Familiar entities

e.g. *Plato's work Republic, Monet's paintings, The Great Pyramids of Giza, The actor Leonardo diCaprio, Canada's maple syrup, ...*

► Subjective adjectives: **positive vs. negative valenced**

e.g. [*inspiring/silly*], [*beautiful/ugly*], [*amazing/boring*], [*extraordinary/overrated*], [*tasty/disgusting*], ...

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e.g. [*inspiring/silly*], [*beautiful/ugly*], [*amazing/boring*], [*extraordinary/overrated*], [*tasty/disgusting*], ...

► Continuation: **True vs. False**

e.g. [...*read by many scholars/unknown among scholars*], [...*largely intact/damaged*], [...*starred in "Titanic"/"Terminator"*], [...*is sweet/bitter*], ...

All items were pre-tested: controversial statements received lower ratings ($M = 2.60$) than non-controversial ($M = 5.32$) and veridical ones ($M = 5.08$) than non-veridicals ($M = 1.42$).

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► Familiar entities

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► [Opinion type: Positive vs. Negative]

e.g. [*inspiring/silly*], [*beautiful/ugly*], [*amazing/boring*], [*extraordinary/overrated*], [*tasty/disgusting*], ...

► [Veridicality: Veridical vs. Non-Veridical]

e.g. [...*read by many scholars/unknown among scholars*], [...*largely intact/damaged*], [...*starred in "Titanic"/"Terminator"*], [...*is sweet/bitter*], ...

Experiment I (stimuli)

► [Packaging: Given], [Positive/Negative], [Veridical/Non-Veridical]

(6a) The [*amazing*[pos]/*boring*[neg]] Great Pyramids of Giza *are largely* [*intact/damaged*].

► [Packaging: New], [Positive/Negative], [Veridical/Non-Veridical]

(6b) “The Great Pyramids of Giza are [*amazing*[pos]/*boring*[neg]] and *largely* [*intact/damaged*].”

Stimuli design (Experiment 2, Baseline)

Stimuli consisted of familiar entities **(i)** presented in definite NPs, combined with **matching** or **not-matching objective adjectives (ii)** and true or false continuations **(iii)**.

(Kaiser & Wang, 2021, p. 20)

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► Familiar entities

e.g. *Plato's work Republic, Monet's paintings, The Great Pyramids of Giza, The actor Leonardo diCaprio, Canada's maple syrup, ...*

► Objective adjectives: Matching vs. Not-Matching Property

e.g. [*ancient/recent*], [*Impressionist/Futurist*], [*Egyptian/Mexican*], [*American/Afro-American*], [*sweet/salty*], ...

(Kaiser & Wang, 2021, p. 20)

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► Continuation: True vs. False

e.g. [...*read by many scholars/unknown among scholars*], [...*largely intact/damaged*], [...*starred in "Titanic"/"Terminator"*], [...*is sweet/bitter*], ...

(Kaiser & Wang, 2021, p. 20)

Experiment 2 (stimuli)

► [Packaging: Given], [Matching/Non-Matching], [Veridical/Non-Veridical]

(7a) The [*American/Afro-American*] actor Leonardo DiCaprio *starred in* [*Titanic/Terminator*].

► [Packaging: New], [Matching/Non-Matching], [Veridical/Non-Veridical]

(7b) “The actor Leonardo DiCaprio is [*American/A-American*] and *starred in* [*Titanic/Terminator*].”

Experiment 1 (stimuli)

► [Packaging: Given], [Positive/Negative], [Veridical/Non-Veridical]

(6a) The [*amazing*^[pos]/*boring*^[neg]] Great Pyramids of Giza *are largely* [*intact/damaged*]

► [Packaging: New], [Positive/Negative], [Veridical/Non-Veridical]

(6b) “The Great Pyramids of Giza are [*amazing*^[pos]/*boring*^[neg]] and *largely* [*intact/damaged*]

Experiment 2 (stimuli)

► [Packaging: Given], [Matching/Non-Matching], [Veridical/Non-Veridical]

(7a) The [*American*/*Afro-American*] actor Leonardo DiCaprio *starred in* [*Titanic/Terminator*]

► [Packaging: New], [Matching/Non-Matching], [Veridical/Non-Veridical]

(7b) “The actor Leonardo DiCaprio is [*American*/*A-American*] and *starred in* [*Titanic/Terminator*]

Experiments 1,2 Overview

Experiments 1,2

Participants

Experiment 1: Subjective

- ▶ Participants: **46/48**
- ▶ Recruitment: **Prolific**
- ▶ Accuracy: 98-100%
- ▶ Language: English (NS)
- ▶ Nationality: USA/UK
- ▶ Location: USA/UK
- ▶ Age: 20-35
- ▶ Reward: £9.5/hour

Experiment 2: Objective

- ▶ Participants: 48/48
- ▶ Recruitment: **Prolific**
- ▶ Accuracy: 98-100%
- ▶ Language: English (NS)
- ▶ Nationality: USA/UK
- ▶ Location: USA/UK
- ▶ Age: 20-35
- ▶ Reward: £ 9.5/hour

In **Experiment 1**, two participants were excluded due to excessively long reading and evaluation times, possibly indicating that they consulted external sources to verify the factual information. Overall median completion times were **15 minutes** for Experiment 1 and **17 minutes** for Experiment 2, within a **50-minute** window.)

Experiments 1,2

Research Question and Hypotheses

Research Question

To what extent does **information packaging** modulate the truth-conditional contribution of **subjective adjectives**?

Experiments 1,2

Research Question and Hypotheses

Hypotheses (Experiment 1)

(H1) **Structure-sensitive at-issue redistribution:** (**Given** $\Rightarrow$ **New shift**)

► Distribution of **False** responses for **subjective adjectives**: (**New**_{Neg}) > (**Given**_{Neg})

(H2) **High Vigilance:** (**Given** $\nRightarrow$ **New shift**)

► Distribution of **False** responses for **subjective adjectives**: (**New**_{Neg}) $\approx$ (**Given**_{Neg})

Baseline (Experiment 2)

(H3) **Truth-conditional invariance** (**New**) $\approx$ (**Given**)

► Distribution of **False** responses for **objective adjectives**: (**New**) $\approx$ (**Given**)

Experiments 1,2

Experimental Design

Experiments 1,2 (Main Trials)

IVs	Packaging	Opinion	Veridicality
Lv 1	Given	Negative	Veridical
Lv 2	New	Positive	Non-veridical
Conditions: 8, Lists: 1-8			

IVs	Packaging	Property	Veridicality
Lv 1	Given	No Match	Veridical
Lv 2	New	Match	Non-veridical
Conditions: 8, Lists: 1-8			

- ▶ **Experiment 1:** Subjective adjectives
- ▶ **Experiment 2:** Objective adjectives
- ▶ 48 Test items
- ▶ 48 Fillers
- ▶ 6 Attention Checks
- ▶ **Method:** offline (Truth-value judgment task)
- ▶ **Design:** 2x2x2 (Within-subjects)
- ▶ **DV:** Truth-judgments responses (True, False)

Experiments 1,2

Experimental Design

Experiments 1,2 (Post-Test)

IVs	Veridicality
Lv 1	<i>Veridical</i>
Lv 2	<i>Non-veridical</i>

- ▶ **Experiment 1:** Subjective adjectives
- ▶ **Experiment 2:** Objective adjectives
- ▶ **48** Test items
- ▶ **6** Attention Checks

Experiments 1,2

...

Rationale for the Post-test

Experiments 1,2

Rationale for the Experiments

Rationale for the Experiments

Since **each stimulus is presented as a conjunction**, the experiment is divided into a **main** phase and a **post-test phase**. The latter will also allow us to exclude who were **not confident participants** about the veridical content.

Experiments 1,2

Rationale for the Experiments

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Since **each stimulus is presented as a conjunction**, the experiment is divided into a **main** phase and a **post-test phase**. The latter will also allow us to exclude who were **not confident participants** about the veridical content.

► Stimuli:

(8a) The [*amazing/boring*] Great Pyramids of Giza *are largely* [*intact/damaged*]"

(8b) "The Great Pyramids of Giza are [*amazing/boring*] and *largely* [*intact/damaged*]"

☐ **True** ☐ **False**

Experiments 1,2

Rationale for the Experiments

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► Stimuli:

(8a) The [*amazing/boring*] Great Pyramids of Giza *are largely* [*intact/damaged*]"

(8b) "The Great Pyramids of Giza are [*amazing/boring*] and *largely* [*intact/damaged*]"

☐ True ☐ False

► Veridical content:

(.c) The GPG *are largely* [*intact/damaged*]

? True ? False

Experiments 1,2

Rationale for the Experiments

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Since **each stimulus is presented as a conjunction**, the experiment is divided into a **main** phase and a **post-test phase**. The latter will also allow us to exclude who were **not confident participants** about the veridical content.

► Stimuli:

(8a) The [*amazing/boring*] Great Pyramids of Giza *are largely* [*intact/damaged*]"

(8b) "The Great Pyramids of Giza are [*amazing/boring*] and *largely* [*intact/damaged*]"

☐ True ☐ False

► Veridical content:

(.c) The GPG *are largely* [*intact/damaged*]

? True ? False

► Opinion-based content:

(.d) The GPG are [*amazing*^[pos]/*boring*^[neg]]

? True ? False

Experiments 1,2

Rationale for the Experiments

Rationale for the Experiments

Since **each stimulus is presented as a conjunction**, the experiment is divided into a **main** phase and a **post-test phase**. The latter will also allow us to exclude who were **not confident participants** about the veridical content.

► Stimuli: (Main Phase)

(8a) The [*amazing/boring*] Great Pyramids of Giza *are largely* [*intact/damaged*]"

(8b) "The Great Pyramids of Giza are [*amazing/boring*] and *largely* [*intact/damaged*]"

☐ True ☐ False

► Veridical content: (Post-Test)

(.c) The GPG *are largely* [*intact/damaged*]

☐ True ☐ False ☐ Don't know

► Opinion-based content: (Likely)

(.d) The GPG are [*amazing*^[pos]/*boring*^[neg]]

☐ True ☐ False

Experiments 1,2

Rationale for the Experiments

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Since **each stimulus is presented as a conjunction**, the experiment is divided into a **main** phase and a **post-test phase**. The latter will also allow us to exclude who were **not confident participants** about the veridical content.

► Stimuli: (Main Phase)

(8a) The [*amazing/boring*] Great Pyramids of Giza *are largely* [*intact/damaged*]"

(8b) "The Great Pyramids of Giza are [*amazing/boring*] and *largely* [*intact/damaged*]"

☐ True ☒ False

► Veridical content: (Post-Test)

(.c) The GPG *are largely* [*intact/damaged*]

☒ True ☐ False ☐ Don't know

► Opinion-based content: (Likely)

(.d) The GPG are [*amazing*^[pos]/*boring*^[neg]]

☐ True ☒ False

Experiments 1,2

Rationale for the Experiments

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Since **each stimulus is presented as a conjunction**, the experiment is divided into a **main** phase and a **post-test phase**. The latter will also allow us to exclude who were **not confident participants** about the veridical content.

► Stimuli: (Main Phase)

(8a) The [*amazing/boring*] Great Pyramids of Giza *are largely* [*intact/damaged*]"

(8b) "The Great Pyramids of Giza are [*amazing/boring*] and largely [*intact/damaged*]"

☐ True ☒ False

► Veridical content: (Post-Test)

(.c) The GPG *are largely* [*intact/damaged*]

☐ True ☐ False ☒ Don't know

**We exclude the data
of not confident responses**

► Opinion-based content:

(.d) The GPG are [*amazing*^[pos]/*boring*^[neg]]

? True ? False

Experiments 1,2

Experimental Design

Experiment I (Test-Items)

- | | |
|---|---------------------------|
| a. The <i>uninteresting</i> Great Pyramids of Giza are <i>largely intact</i> . | [Given/Neg/Veridical] |
| b. The Great Pyramids of Giza are <i>uninteresting</i> and <i>largely intact</i> . | [New/Neg/Veridical] |
| c. The <i>brehtaking</i> Great Pyramids of Giza are <i>largely intact</i> . | [Given/Pos/Veridical] |
| d. The Great Pyramids of Giza are <i>brehtaking</i> and <i>largely intact</i> . | [New/Pos/Veridical] |
| e. The <i>uninteresting</i> Great Pyramids of Giza are <i>largely damaged</i> . | [Given/Neg/Non-Veridical] |
| f. The Great Pyramids of Giza are <i>uninteresting</i> and <i>largely damaged</i> . | [New/Neg/Non-Veridical] |
| g. The <i>brehtaking</i> Great Pyramids of Giza are <i>largely damaged</i> . | [Given/Pos/Non-Veridical] |
| h. The Great Pyramids of Giza are <i>brehtaking</i> and <i>largely damaged</i> . | [New/Pos/Non-Veridical] |

Instructions



“The uninteresting Great Pyramids of Giza are largely intact.”

True

False

Experiments 1,2

Experimental Design

Experiments I (Post-Test Items)

- (a) The Great Pyramids of Giza are [*largely intact*]. (Veridical)
(b) The Great Pyramids of Giza are [*largely damaged*]. (Non-Veridical)

Instructions



“The Great Pyramids of Giza are largely intact.”

True

False

Don't Know

Experiments 1,2

Experimental Design

Experiment 2 - Baseline (Test-Items)

- | | |
|--|--------------------------------|
| a. The <i>Mexican</i> Great Pyramids of Giza are <i>largely intact</i> . | [Given/No-Match/Veridical] |
| b. The Great Pyramids of Giza are <i>Mexican</i> and <i>largely intact</i> . | [New/No-Match/Veridical] |
| c. The <i>Egyptian</i> Great Pyramids of Giza are <i>largely intact</i> . | [Given/Match/Veridical] |
| d. The Great Pyramids of Giza are <i>Egyptian</i> and <i>largely intact</i> . | [New/Match/Veridical] |
| e. The <i>Mexican</i> Great Pyramids of Giza are <i>largely damaged</i> . | [Given/No-Match/Non-Veridical] |
| f. The Great Pyramids of Giza are <i>Mexican</i> and <i>largely damaged</i> . | [New/No-Match/Non-Veridical] |
| g. The <i>Egyptian</i> Great Pyramids of Giza are <i>largely damaged</i> . | [Given/Match/Non-Veridical] |
| h. The Great Pyramids of Giza are <i>Egyptian</i> and <i>largely damaged</i> . | [New/Match/Non-Veridical] |

Instructions



“The Mexican Great Pyramids of Giza are largely intact.”

True

False

Experiments 1,2

Experimental Design

Experiments I (Post-Test Items)

- (a) The Great Pyramids of Giza are [*largely intact*]. (Veridical)
(b) The Great Pyramids of Giza are [*largely damaged*]. (Non-Veridical)

Instructions



“The Great Pyramids of Giza are largely intact.”

True

False

Don't Know

Experiments 1,2

...

Procedure

Experiments 1,2

Instructions and procedure

TI

“The uninteresting Great Pyramids of Giza are largely intact.”

- ▶ Started sentence-reading RT recording.

Figure 3. “Main Trial” procedure

Experiments 1,2

Instructions and procedure

T1

“The uninteresting Great Pyramids of Giza are largely intact.”

T2

“The uninteresting Great Pyramids of Giza are largely intact.”

True

False

- ▶ **Displayed the instruction:** “Do you consider the following sentence *True* or *False*”.
- ▶ Stopped sentence-reading RT recording.
- ▶ Started evaluation-time ET recording.

Figure 3. “Main Trial” procedure

Experiments 1,2

Instructions and procedure

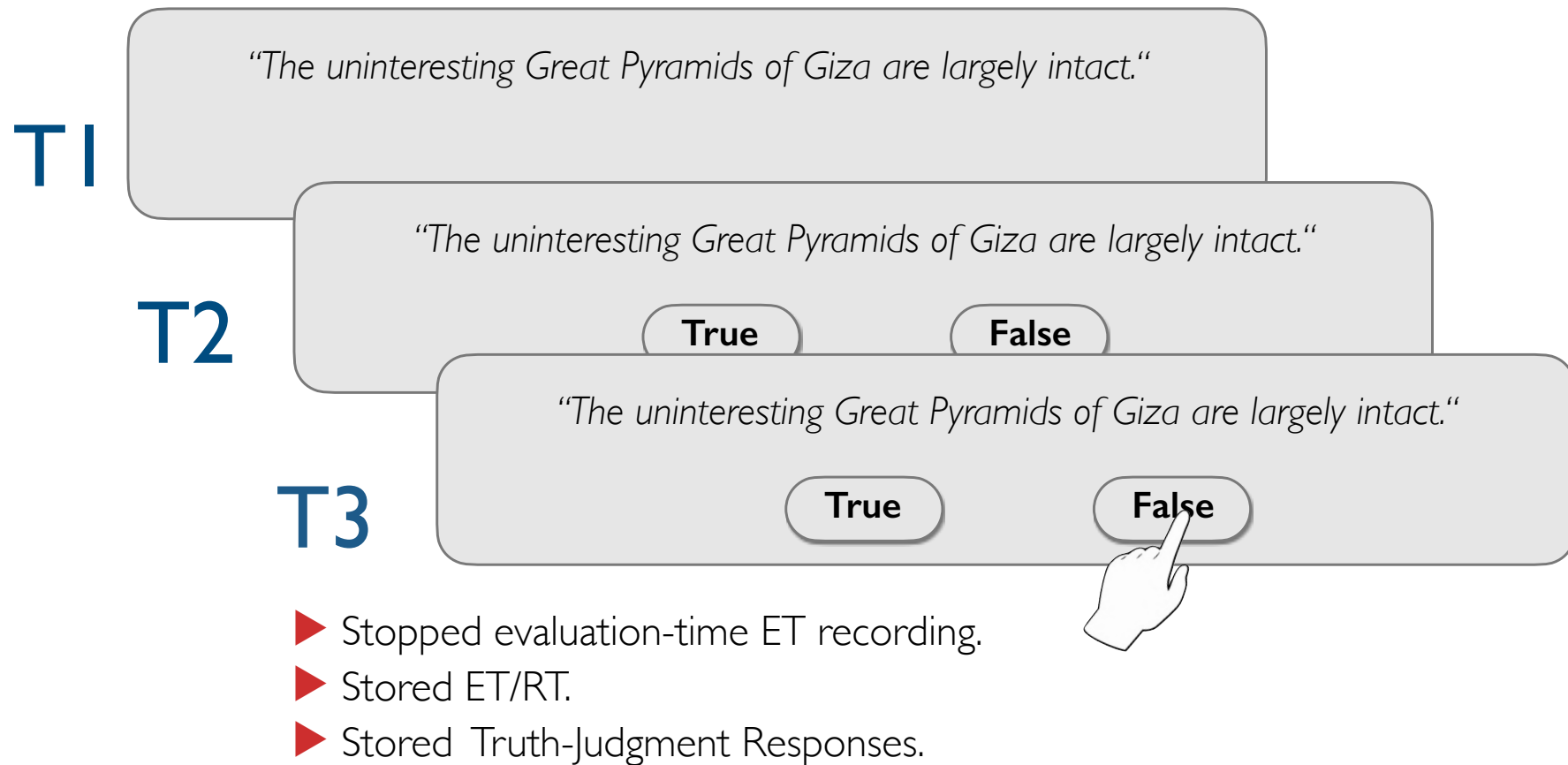


Figure 3. "Main Trial" procedure

Experiments 1,2

Instructions and procedure

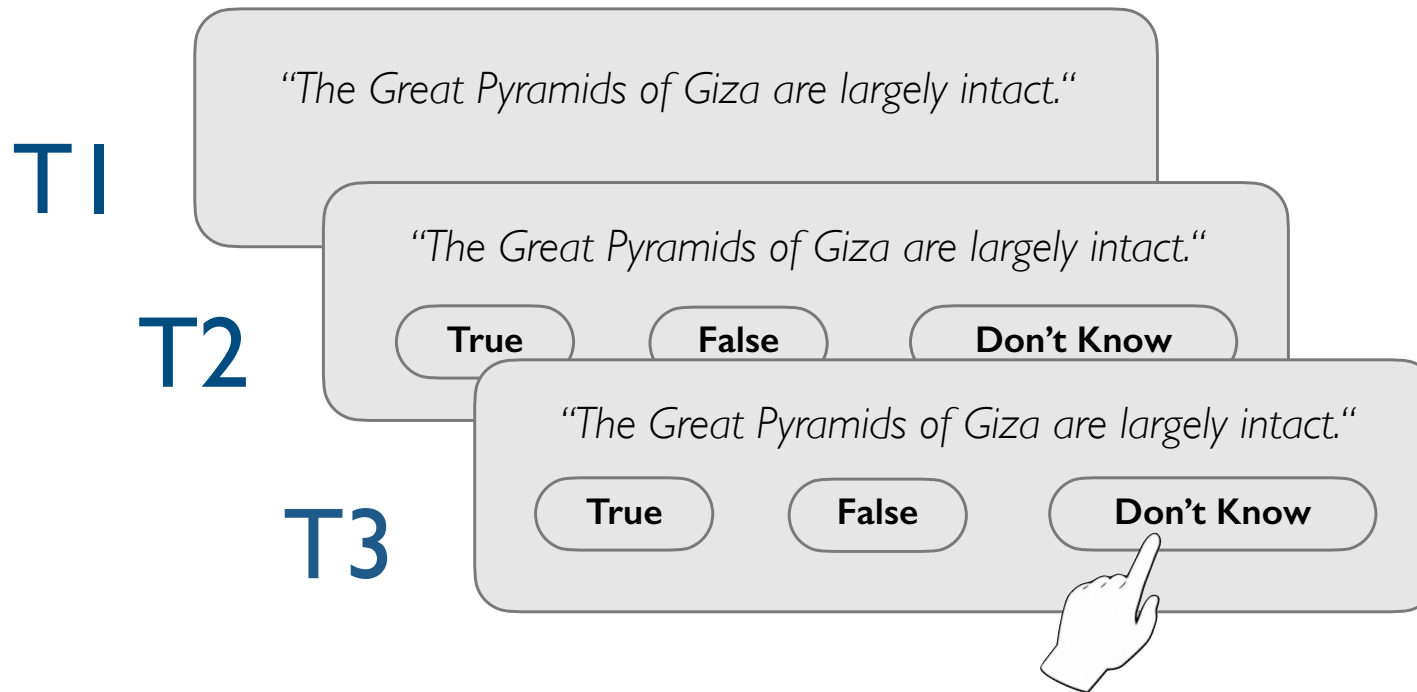


Figure 4. "Post-Test" procedure

Experiments 1,2

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Data Analysis and Results

Mixed-effects modelling

- ▶ Continuous outcomes were analysed with linear mixed-effects models (**LMMs**).
- ▶ Binary TRUE/FALSE responses were analysed with binomial generalized linear mixed-effects models (**GLMMs**). All predictors were deviation-coded.
- ▶ Following Gelman and Hill (2006), effects were considered meaningful when:
 - $|z| > 2$. for truth-judgment responses;
 - $|t| > 2$. for evaluation time (EV) responses;
- ▶ Analyses were conducted in R-Studio (**lme4 package**).

Experiments 1,2

Data and Results

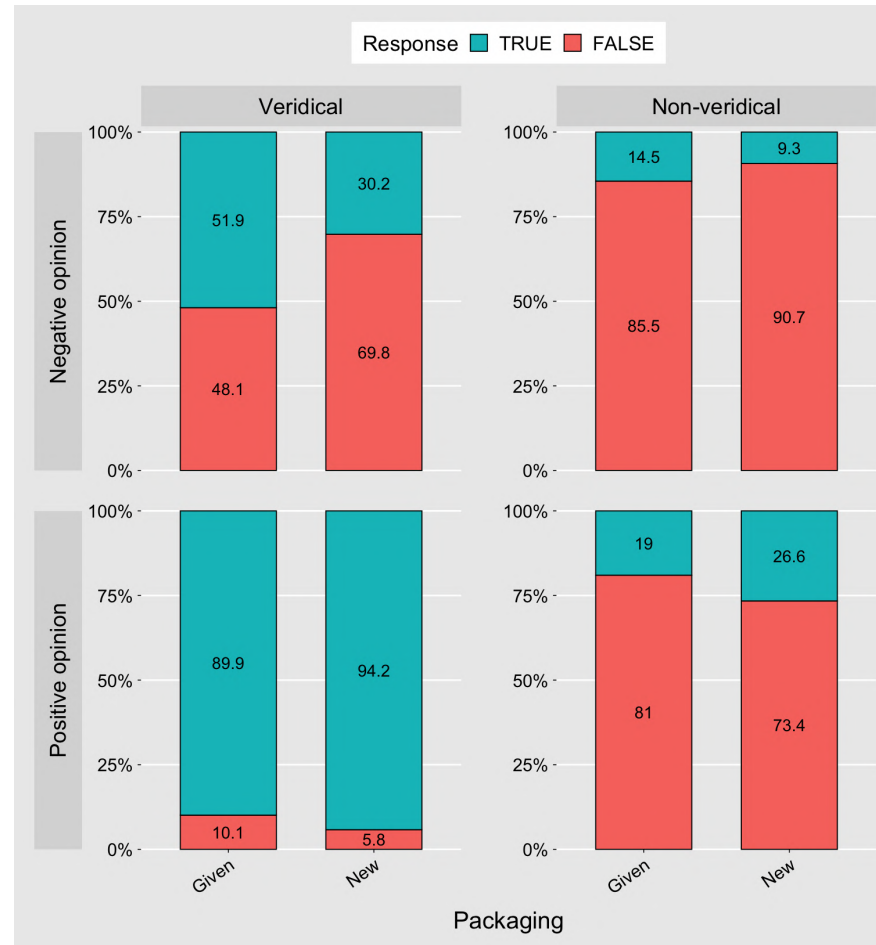


Figure 5. Mean distribution of TRUE and FALSE responses across conditions in **Experiment I**.

Experiments 1,2

Data and Results

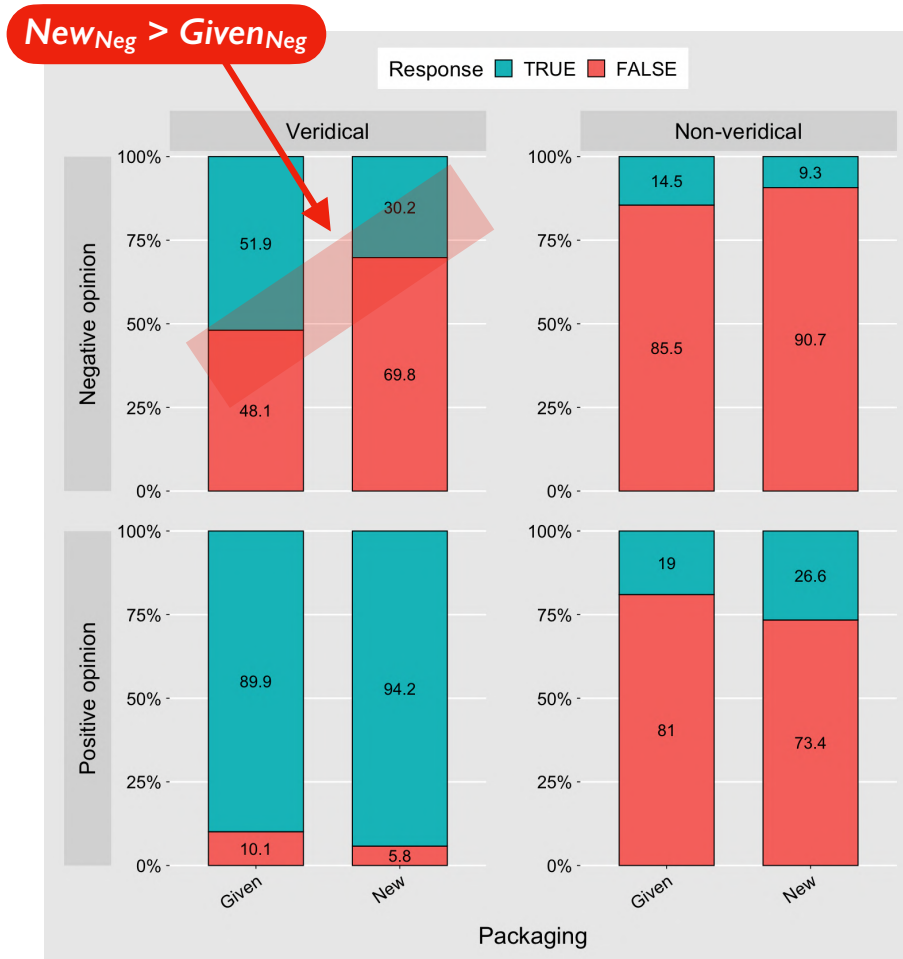


Figure 5. Mean distribution of TRUE and FALSE responses across conditions in **Experiment I**.

▶ The [*amazing/boring*] GPG [*are largely intact/damaged*]

▶ The GPG are [*amazing/boring*] and [*largely intact/*

(H1) Structure-sensitive at-issue distribution ✓

▶ False responses: **New_{Neg} > Given_{Neg}**

(H2) No packaging-driven at-issue shift ✗

▶ False responses: **New ≈ Given**

Fixed effects	Estimate	SE	z value
Structure	-0.194	0.070	-2.752
Opinion type	-0.676	0.071	-9.535
Veridicality	1.348	0.075	17.999
Structure:Opinion type	-0.318	0.070	-4.513
Opinion type:Veridicality	-0.277	0.071	-3.929

Experiments 1,2

Data and Results

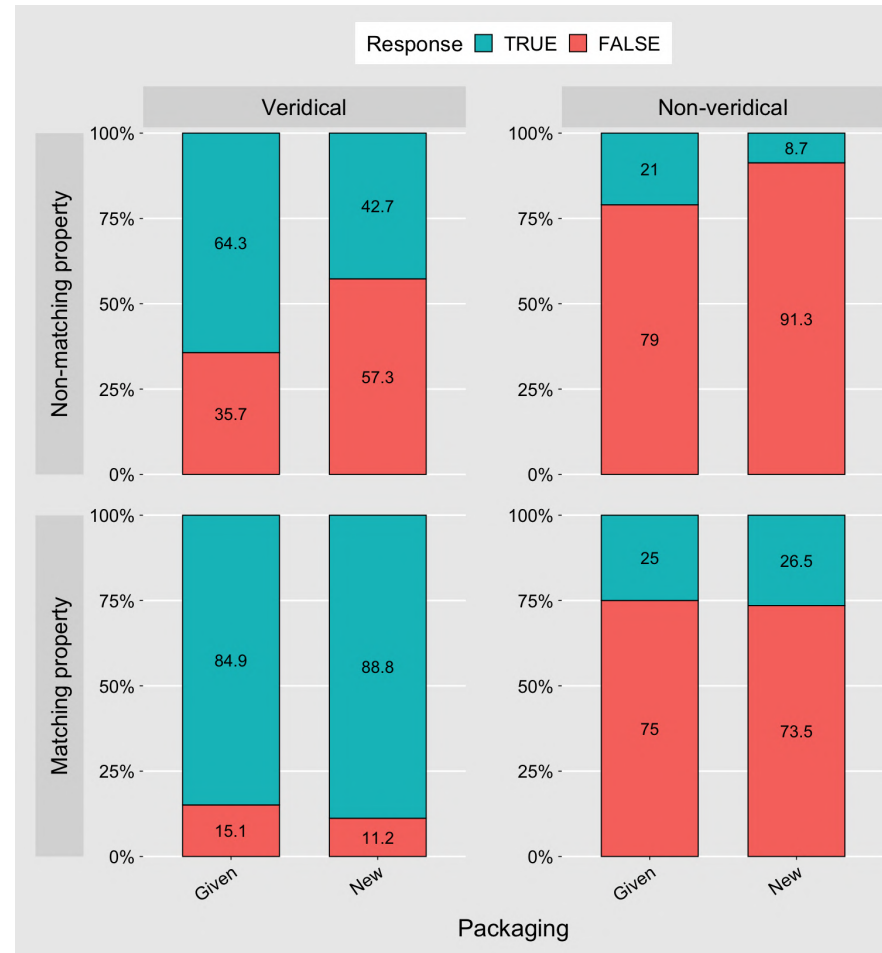


Figure 6. Mean distribution of TRUE and FALSE responses across conditions in **Experiment 2**.

Experiments 1,2

Data and Results



Figure 6. Mean distribution of TRUE and FALSE responses across conditions in **Experiment 2**.

- ▶ The [*Mexican/Egyptian*] GPG [*are ... intact/damaged*]
- ▶ The GPG are [*Mexican/Egyptian*] and [*intact/damaged*]

(H3) No packaging-driven at-issue shift ✗

- ▶ False responses: **$New_{Mis} \approx Given_{Mis}$**

Fixed effects	Estimate	SE	z value
Structure	0.200	0.066	-3.044
Opinion type	0.674	0.069	-9.798
Veridicality	1.321	0.073	18.165
Structure:Opinion type	-0.297	0.066	-4.502
Opinion type:Veridicality	0.273	0.068	4.005

Experiments 1,2

Data and Results

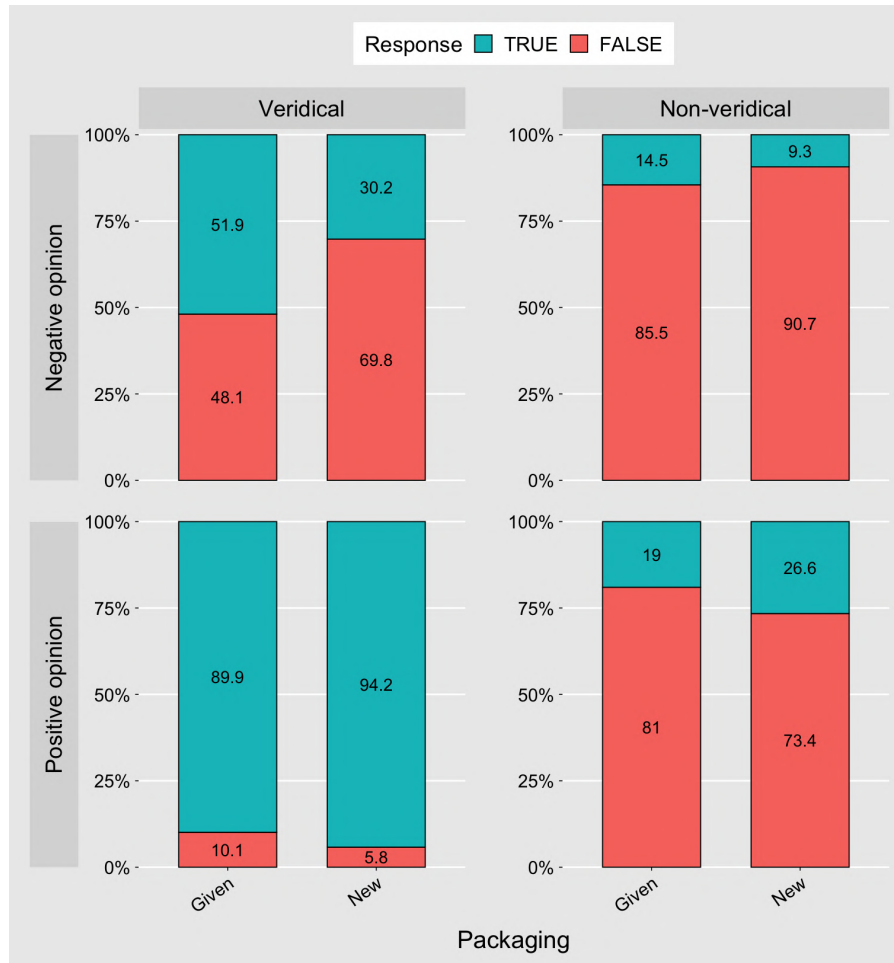


Figure 5. Mean distribution of TRUE and FALSE responses across conditions in **Experiment 1**.

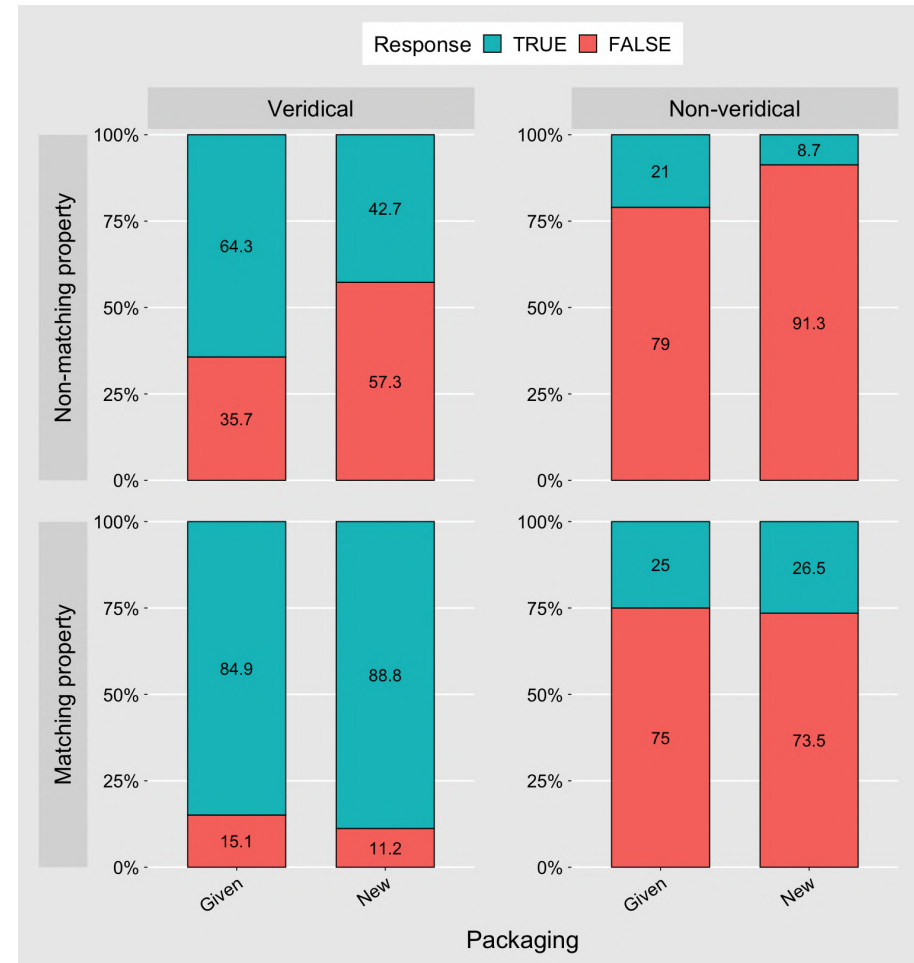


Figure 6. Mean distribution of TRUE and FALSE responses across conditions in **Experiment 2**.

Experiments 1,2

Data and Results

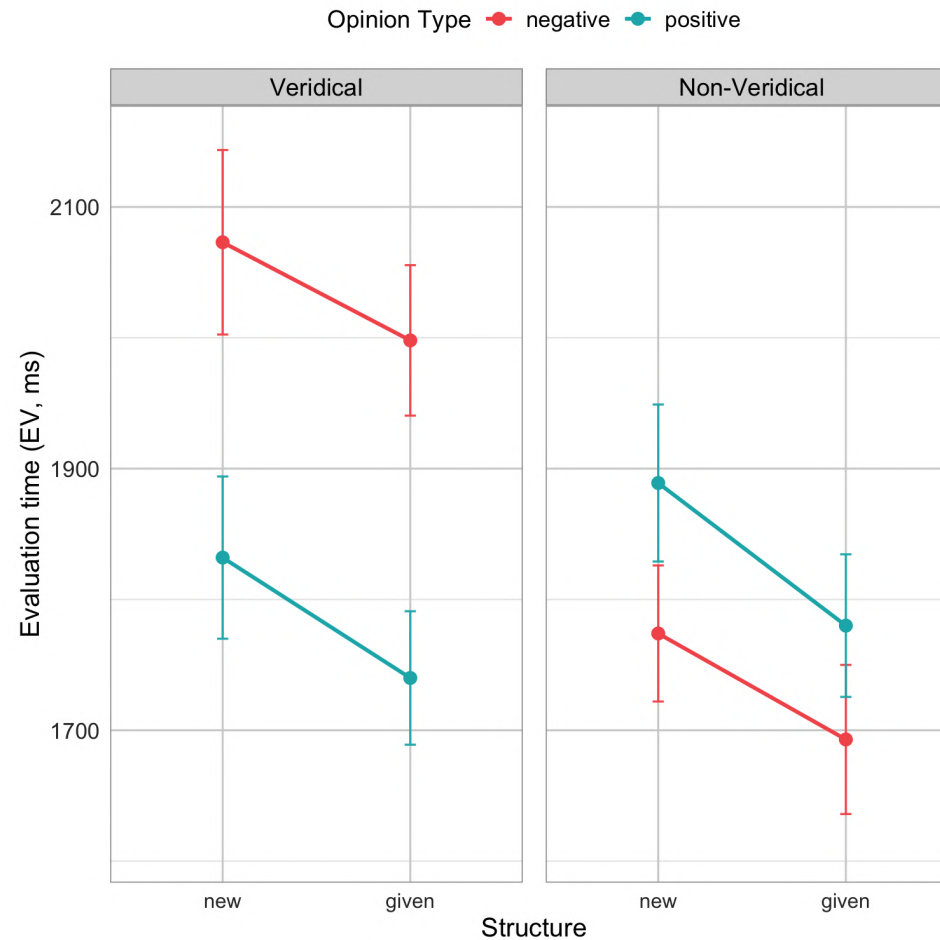


Figure 7. Evaluation Time (EV) in ms. error bars indicate ± 1 standard deviation in **Experiment 1**.

Experiments 1,2

Data and Results

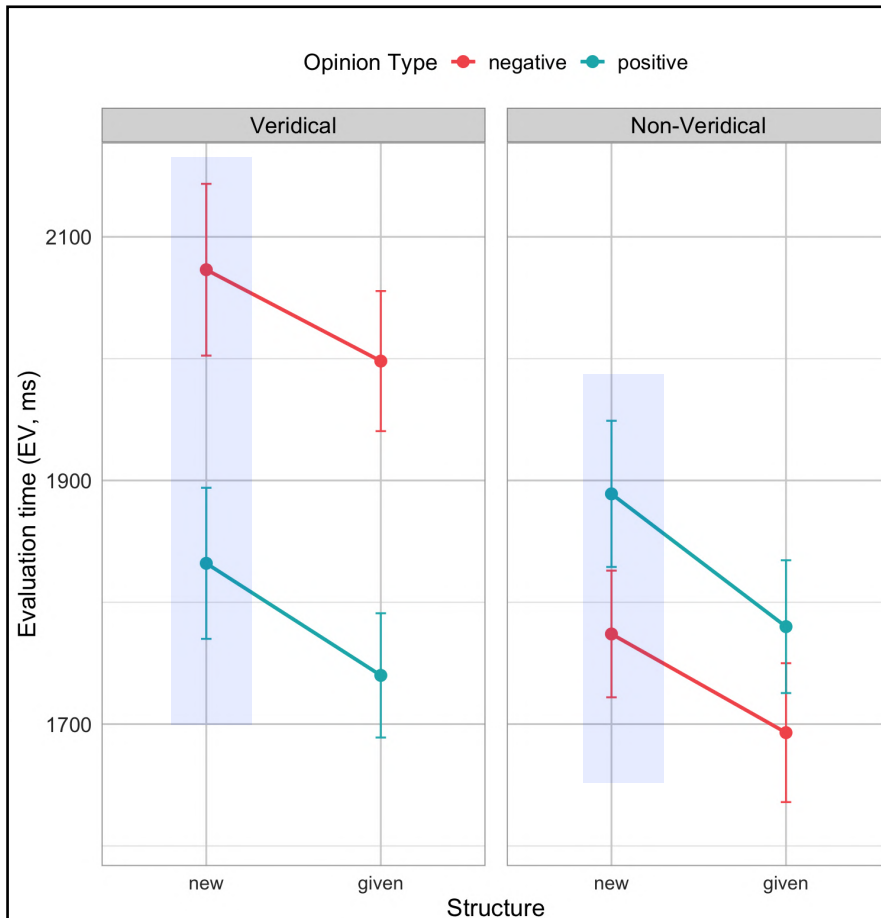


Figure 7. Evaluation Time (EV) in ms. error bars indicate ± 1 standard deviation in **Experiment I**.

- a. The [*amazing/boring*] GPG [*are largely intact/damaged*]
- b. The GPG are [*amazing/boring*] and [*a.l. intact/damaged*]

EV Time (for LMMs) ✓ interaction

- ▶ When Veridical: **Negative** > **Positive**
- ▶ When Veridical: **New_{Neg}** > **New_{Pos}**
- ▶ When Non-Veridical: **New_{Pos}** > **New_{Neg}**

Possible guess: Participants seem sensitive to **the opinion-based** and **factual** like **propositions** introduced as **new information**, especially when their **truth values diverge**, making it unclear which should weigh more in the response. ()

Experiments 1,2

Data and Results

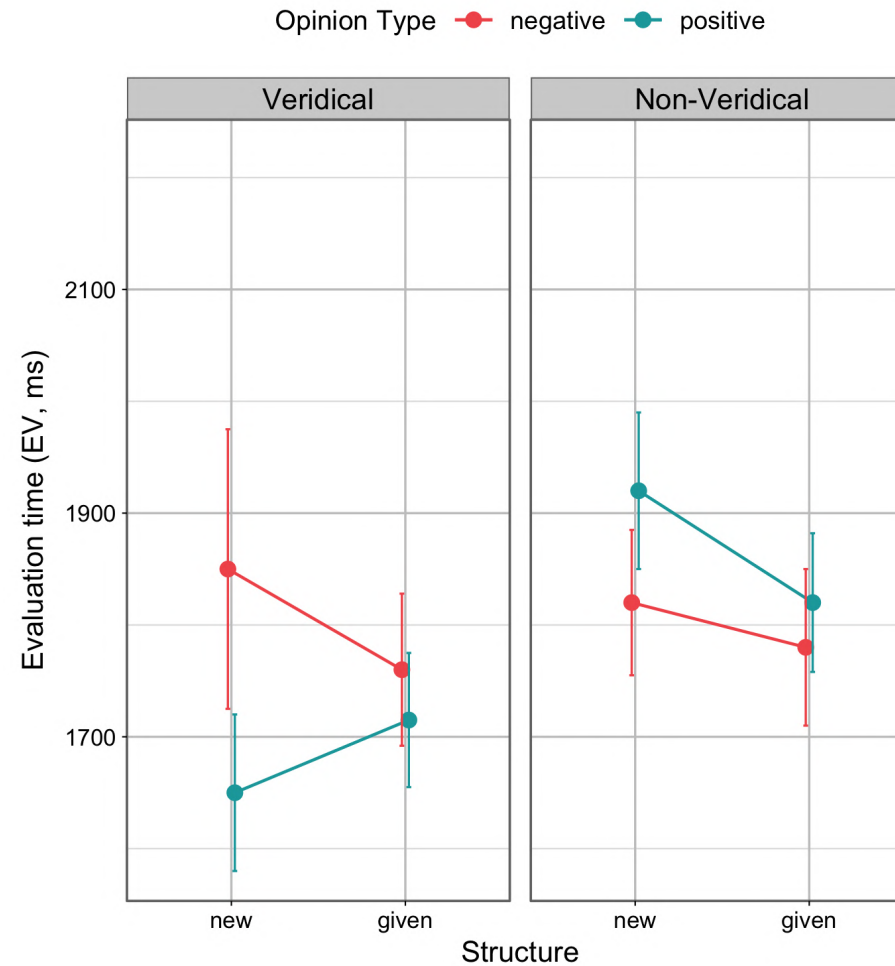


Figure 8. Evaluation Time (EV) in ms. error bars indicate ± 1 standard deviation in **Experiment 2**.

Experiments 1,2

Data and Results

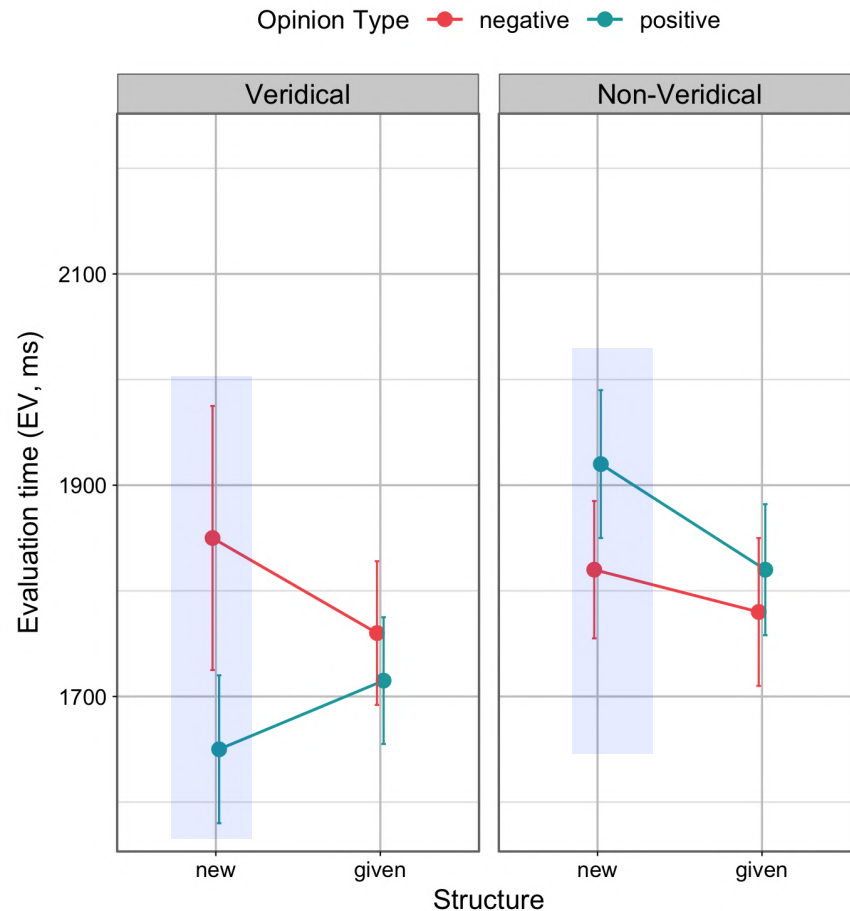


Figure 8. Evaluation Time (EV) in ms. error bars indicate ± 1 standard deviation in **Experiment 2**.

- a. The [*amazing/boring*] GPG [*are largely intact/damaged*]
- b. The GPG are [*amazing/boring*] and [*a.l. intact/damaged*]

EV Time (for LMMs) $\times$ not significant

- ▶ When Veridical: **Negative > Positive**
- ▶ When Veridical: **New_{Neg} > New_{Pos}**
- ▶ When Non-Veridical: **New_{Pos} > New_{Neg}**

Possible guess: As in Experiment 1, participants seem sensitive to competing propositions, but **when the information is clearly false or misleading, decision making is faster.**

Experiments 1,2

Data and Results

General Findings

Experiments 1,2

Data and Results

General Findings

► Overall structural packaging effects

Experiments 1 and **2** replicate the structural patterns found by **Kaiser & Wang (2021)** and **Delogu & Degaetano-Ortlieb (forthcoming)**.

Experiments 1,2

Data and Results

General Findings

► Overall structural packaging effects

Experiments 1 and 2 replicate the structural patterns found by **Kaiser & Wang (2021)** and **Delogu & Degaetano-Ortlieb (forthcoming)**.

► The impact of packaging on the semantic dimension

Experiments 1 and 2 reveal clear structural effects: **negatively valenced opinions** receive more TRUE responses when presented structurally as **Given** rather than **New information**.

Experiments 1,2

Data and Results

General Findings

► Overall structural packaging effects

Experiments 1 and 2 replicate the structural patterns found by **Kaiser & Wang (2021)** and **Delogu & Degaetano-Ortlieb (forthcoming)**.

► The impact of packaging on the semantic dimension

Experiments 1 and 2 reveal clear structural effects: **negatively valenced opinions** receive more TRUE responses when presented structurally as **Given** rather than **New information**.

► **Attention allocation and the unexpected baseline result**

A surprising pattern emerged for **the baseline condition** involving objective adjectives in Experiment 2. Do comprehenders partly overlook information depending on where it is structurally presented?

Experiments 1,2

Data and Results

General Findings

► Overall structural packaging effects

Experiments 1 and 2 replicate the structural patterns found by **Kaiser & Wang (2021)** and **Delogu & Degaetano-Ortlieb (forthcoming)**.

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Experiments 1 and 2 reveal clear structural effects: **negatively valenced opinions** receive more TRUE responses when presented structurally as **Given** rather than **New information**.

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A surprising pattern emerged for **the baseline condition** involving objective adjectives in Experiment 2. Do comprehenders partly overlook information depending on where it is structurally presented?

► **Positive and Negative Valence**

Packaging effects were observed for negative, but not positive, content. Why?

Experiments 1,2

Data and Results

General Findings

► Overall structural packaging effects

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Experiments 1 and 2 reveal clear structural effects: **negatively valenced opinions** receive more TRUE responses when presented structurally as **Given** rather than **New information**.

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A surprising pattern emerged for **the baseline condition** involving objective adjectives in Experiment 2. Do comprehenders partly overlook information depending on where it is structurally presented?

► Positive and Negative Valence

Packaging effects were observed for negative, but not positive, content. Why?

Experiments 1,2

...

Final Considerations

Attention Allocation	Topic/Given	Comment/New
<i>Vigilance</i>	<i>Low</i>	<i>High</i>
<i>Egocentric bias</i>	<i>Sale resistance (low)</i>	<i>Sale resistance (high)</i>
<i>Force</i>	<i>Locutionary Force</i>	<i>Illocutionary Force</i>
<i>Dual processing</i>	<i>Automatic processing</i>	<i>Controlled Processing</i>
<i>RTs/Cog.effort</i>	<i>Faster/Lower</i>	<i>Slower/Higher</i>

Table 1. Expected cognitive processing effects as a function of the information structure of the sentence.

(Rosengren, 1992, 1993; Ferreira et al., 2002; Ward & Sturt, 2007; Sperber et al., 2010; Christiansen & Chater, 2016; Lombardi Vallauri, 2016; Domaneschi et al., 2018; Lombardi Vallauri, 2021)

Attention Allocation Account (AAA) vs. Expressive-based Account (EBA)

AAA predicts that attention targets the **New** content, while the opinion-based content in **(9b)** is accommodated. By contrast, **EBA** treats the opinion-based content as a **speaker-oriented proposition**: what enters the common ground is **(10c)**, and would be odd to reject it since it is a subjective proposition.

► Attention-Allocation Account (AAA)

(9a) “Plato’s *silly* Republic is read by many scholars.”

(9b) Given: Plato’s Republic is *silly*[**neg**]

(9c) New: Plato’s Republic is read by many scholars.

(Low attention/Vigilance)

(High attention/Vigilance)

Attention Allocation Account (AAA) vs. Expressive-based Account (EBA)

AAA predicts that attention targets the **New** content, while the opinion-based content in (9b) is accommodated. By contrast, **EBA** treats the opinion-based content as a **speaker-oriented proposition**: what enters the common ground is (10c), and would be odd to reject it since it is a subjective proposition.

► Attention-Allocation Account (AAA)

(9a) “Plato’s silly Republic is read by many scholars.”

(9b) **Given:** Plato’s Republic is silly^[neg]

(Low attention/Vigilance)

(9c) **New:** Plato’s Republic is read by many scholars.

(High attention/Vigilance)

► Expressive-based Account (EBA)

(10a) “Plato’s silly Republic is read by many scholars.”

(10b) **Given:** Plato’s Republic is silly^[neg] to the speaker.

(Speaker-based evaluative content*)

(10c) **New:** Plato’s Republic is read by many scholars.

(Factual content)

Attention Allocation Account (AAA) vs. Expressive-based Account (EBA)

AAA predicts that attention targets the **New** content, while the opinion-based content in **(9b)** is accommodated. By contrast, **EBA** treats the opinion-based content as a **speaker-oriented proposition**: what enters the common ground is **(10c)**, and would be odd to reject it since it is a subjective proposition.

► Attention-Allocation Account (AAA)

(9a) “Plato’s *silly* Republic is read by many scholars.”

(9b) Given: Plato’s Republic is *silly*[**neg**]

(**Low** attention/Vigilance)

(9c) New: Plato’s Republic is read by many scholars.

(**High** attention/Vigilance)

► Expressive-based Account (EBA)

(10a) “Plato’s *silly* Republic is read by many scholars.”

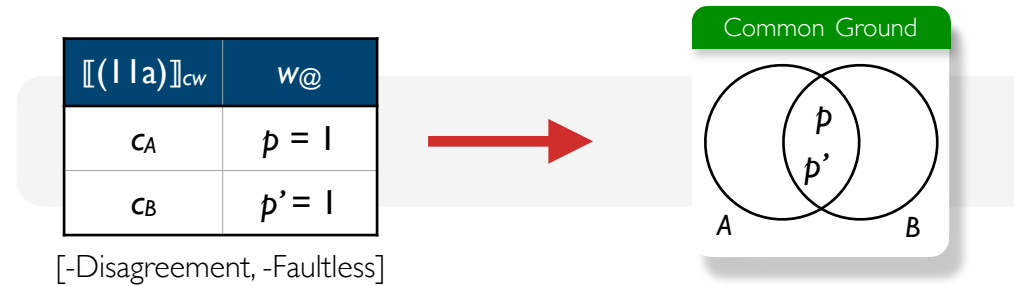
(10b) Given: Plato’s Republic is *silly*[**neg**] to the speaker.

(**Speaker-based evaluative content***)

(10c) New: Plato’s Republic is read by many scholars.

(**Factual content**)

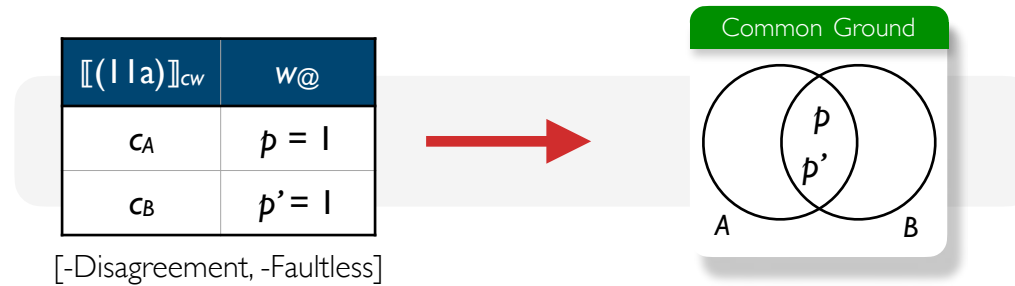
(I Ia) **Reduced pressure:** [Plato's *silly* Republic]^{GIVEN} [is read by many scholars.]^{NEW}



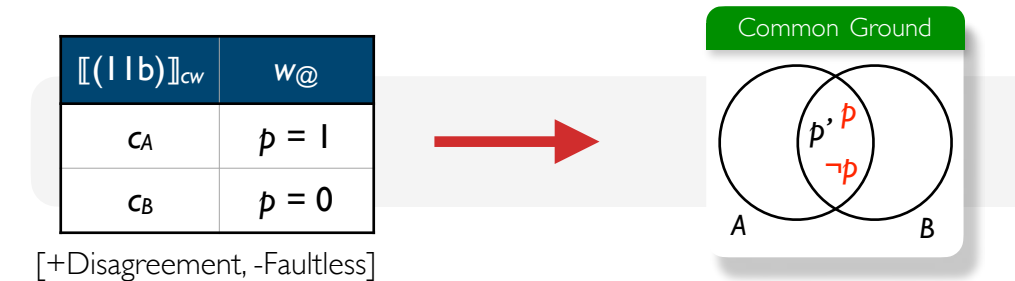
p'

- ▶ $p = \{\text{'Plato's Republic is read by many scholars'}\}$: **factual content**
- ▶ $p' = \{\text{'Plato's Republic is silly to the speaker in } c_w\}$: **Speaker-based evaluative content**

(I Ia) **Reduced pressure:** [Plato's *silly* Republic]^{GIVEN} [is read by many scholars.]^{NEW}



(I Ib) **CG-update pressure:** [Plato's Republic]^{GIVEN} [is *silly* and read by many scholars.]^{NEW}

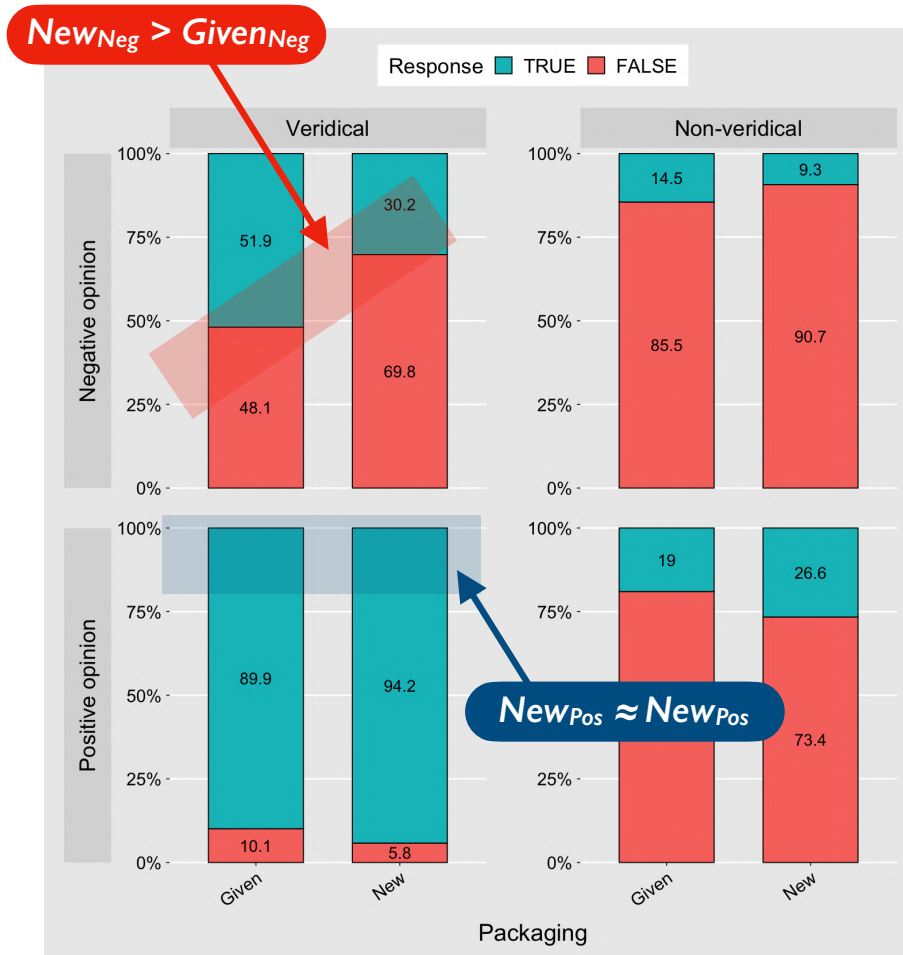


- ▶ $p = \{\text{'Plato's Republic is read by many scholars'}\}$: **factual content**
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Implications for Valence?

Final Considerations

Implications for Valence



Valence asymmetry

Figure 5. Mean distribution of TRUE and FALSE responses across conditions in **Experiment I**.

Final Considerations

Implications for Valence

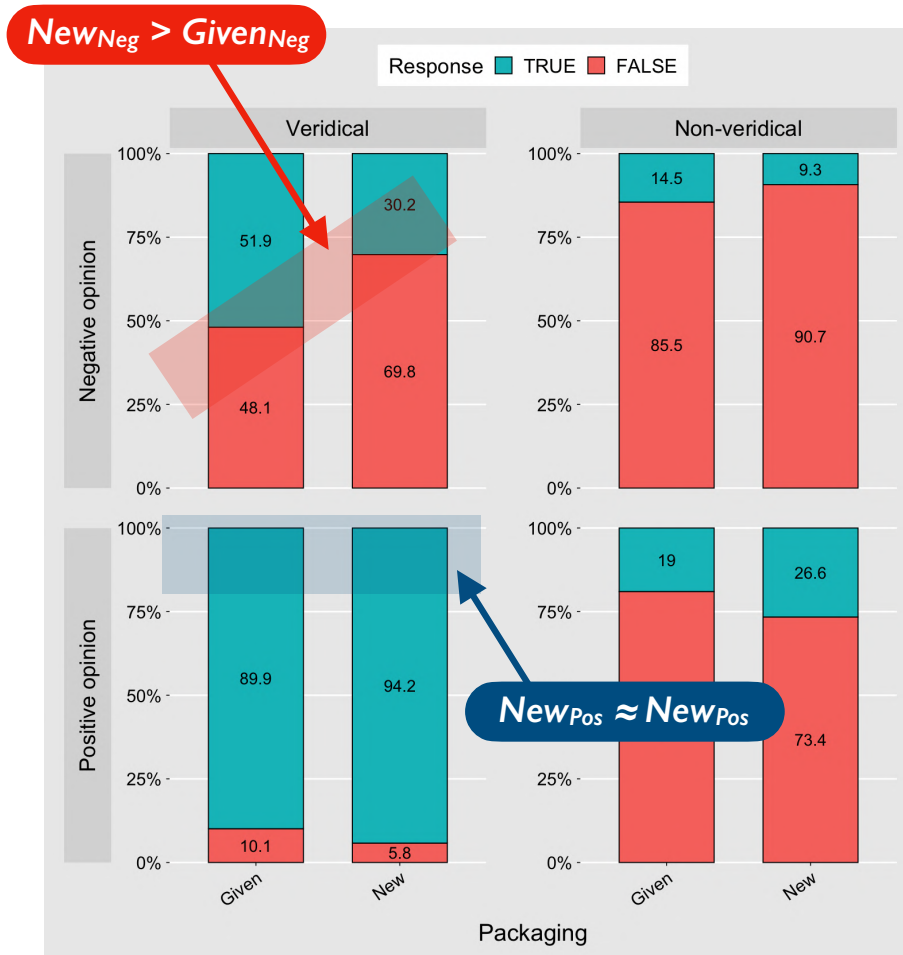


Figure 5. Mean distribution of TRUE and FALSE responses across conditions in **Experiment I**.

Valence asymmetry

► Packaging:

no structural effects for **positive evaluatives**.

Final Considerations

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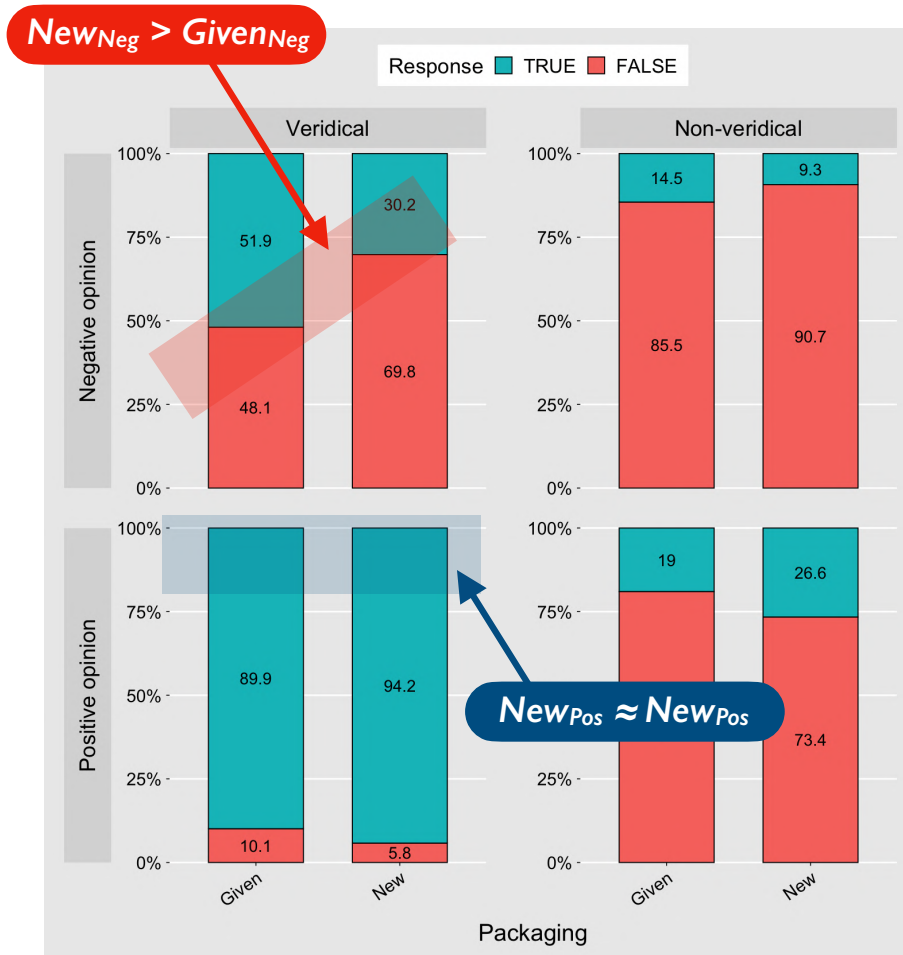


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positive words are more frequent and less marked (Augustine et al. 2011; Warriner et al. 2013).

Final Considerations

Implications for Valence

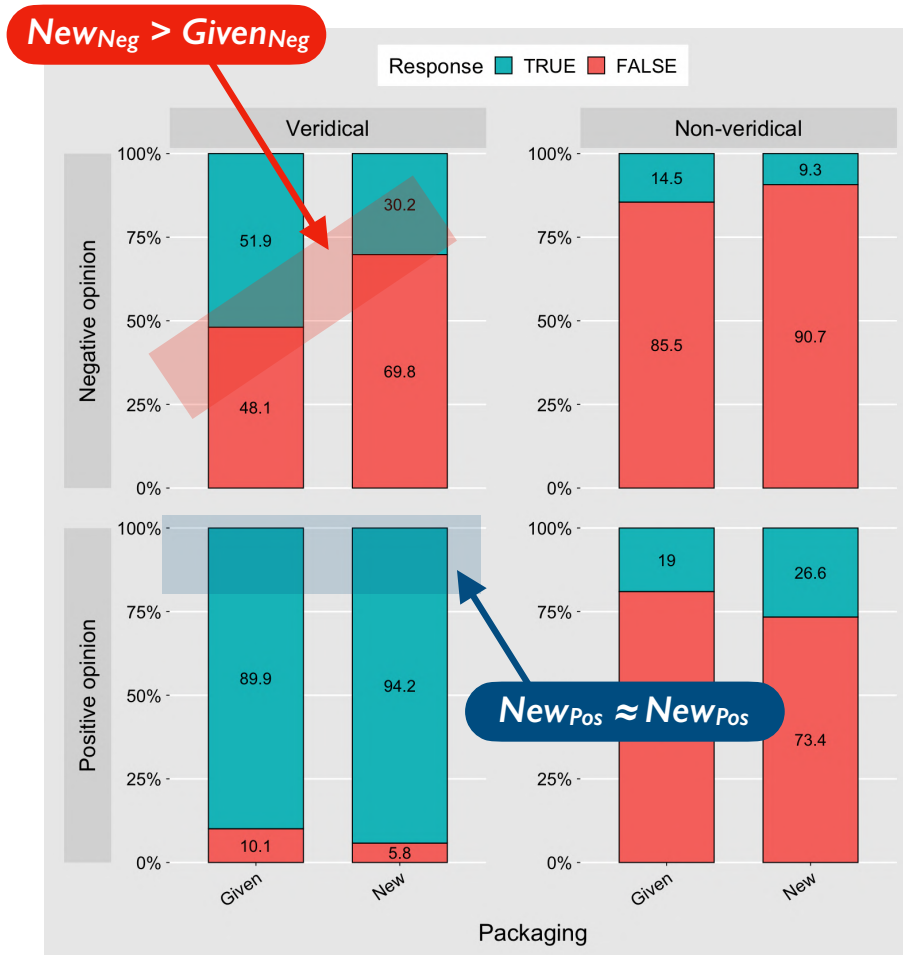


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negative evaluatives demand more attention and processing effort (Alves et al., 2017).

Final Considerations

Implications for Valence

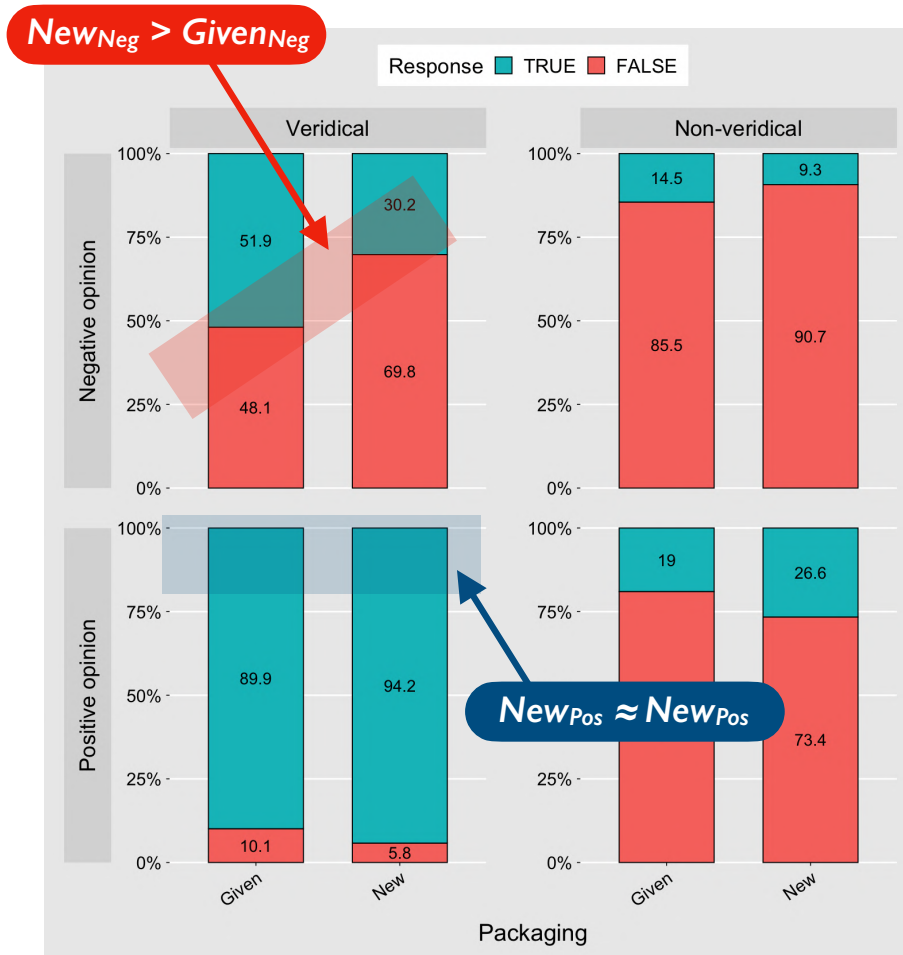


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negative evaluatives demand more attention and processing effort (**Alves et al., 2017**).

► Persuasion:

positive content may function as a subtler persuasive tool (cf. **Kaiser & Wang, 2021; Delogu & Degaetano-Ortlieb 2025**).

That's all! Thank you



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✓ **Quel puttaniere del duce Mussolini viene spesso citato nei libri di storia insieme ad Adolf Hitler.**

☐ sì ☒ **no** → *Mussolini e Hitler* sono effettivamente spesso citati insieme nei libri di storia, ma l'epiteto "puttaniere" è offensivo e non è un fatto storico accertato né pertinente.

✓ **Quella puttana di Giorgia Meloni è l'attuale primo ministro italiano.**

☒ sì (fatto: Giorgia Meloni è l'attuale primo ministro italiano), ma: l'uso dell'epiteto è **gravemente offensivo** e rende la frase inaccettabile in un contesto neutro o didattico.

✓ **Berlusconi era una merda e primo ministro italiano tra gli anni 90-2000.**

☐ sì ☒ **no** → Berlusconi è stato primo ministro negli anni '90 e 2000; "una merda" è insulto e non fatto. La frase non è valutabile come vera in senso fattuale.

✓ **Quella cagna della Schlein è segretaria del PD.**

☒ sì (fatto: Elly Schlein è segretaria del PD), ma: "cagna" è **insulto pesante**, non adatto a una frase valutabile come vera o neutra.

Figure 6. Observed model behavior on prompts containing slurs/pejoratives when answering whether each statement is true or false: the model preserves factual verification but consistently flags offensive language as unacceptable, and it responds "false" only when the target entities are male (sexist bias?). **Italian**

✓ That whoremonger, the Duce Mussolini, is often mentioned in history books together with Adolf Hitler.

☐ yes ☒ no → *Mussolini and Hitler are indeed often mentioned together in history books, but the epithet "whoremonger" is offensive and is neither a verified historical fact nor relevant.*

✓ That whore Giorgia Meloni is the current Italian prime minister.

☒ yes (*fact: Giorgia Meloni is the current Italian prime minister*), but: the use of the epithet is gravely offensive and makes the sentence unacceptable in a neutral or educational context.

✓ Berlusconi was a piece of shit and Italian prime minister between the 1990s and 2000s.

☐ yes ☒ no → Berlusconi was prime minister in the '90s and 2000s; "a piece of shit" is an insult and not a fact. The sentence cannot be evaluated as true in a factual sense.

✓ That bitch Schlein is the secretary of the Democratic Party.

☒ yes (*fact: Elly Schlein is secretary of the PD*), but: "bitch" is a strong insult, not suitable for a sentence that can be evaluated as true or neutral.

Figure 7. Observed model behavior on prompts containing slurs/pejoratives when answering whether each statement is true or false: the model preserves factual verification but consistently flags offensive language as unacceptable, and it responds "false" only when the target entities are male (sexist bias?). **English**